Interior Gateway IP Routing Protocols

Redes de Comunicações II

Licenciatura em Engenharia de Computadores e Informática

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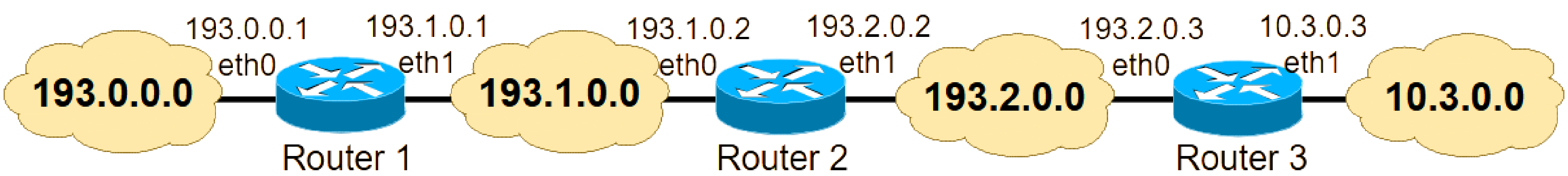
IP Routing Overview

- Routers must know how to forward packets to any existing IP network.
  - A router immediately knows the IP networks directly connected to each of its interfaces
  - For remote IP networks (i.e., IP networks not directly connected to one of its interfaces), the router must rely on additional information.
- In the following example (considering all network of class C):



If no routing is configured:

- Router 1, for example, knows how to reach 193.0.0.0/24 and 193.1.0.0/24 (the directly connected IP networks), but does not know how to reach neither 193.2.0.0/24 nor 10.3.0.0/24

IP Routing Overview

A router can be made aware of remote IP networks through:

- **Static routing**: The routing path from each router to each remote IP network is administratively configured in each router.
- **Dynamic routing protocols**: Routers communicate between them through a protocol to learn how to forward packets towards each remote IP network.
- **Policy based routing**: Additional routing rules (administratively configured in the routers) to define routing policies other than the basic static/dynamic IP routing.

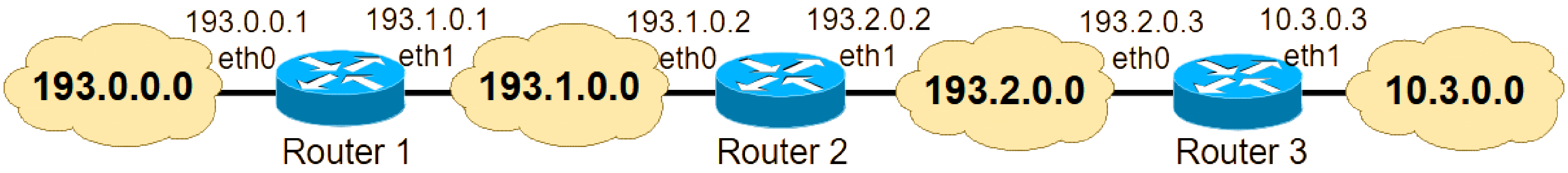
IP Default Routes

- In many cases, a router does not need to recognize each individual remote IP network.
- The router can be configured to send IP packets towards another router for all unknown IP networks (i.e., all IP networks that are not in the IP routing table of the router).
  - This is known as a default route.
- Default routes can be configured either by static routing or using dynamic routing protocols.

  IPv4 default route - 0.0.0.0/0

  IPv6 default route - ::/0

Static Routing Examples



- Router 2 does not know networks 193.0.0.0/24 and 10.3.0.0/24
  - To reach full connectivity, it must be configured with the static routes:
    - 193.0.0.0/24 accessible via 193.1.0.1 (eth1, Router 1)
    - 10.3.0.0/24 accessible via 193.2.0.3 (eth0, Router 3)
- Router 1 does not know networks 193.2.0.0/24 and 10.3.0.0/24
  - To reach full connectivity, it must be configured with the static routes:
    - 193.2.0.0/24 accessible via 193.1.0.2 (eth0, Router 2)
    - 10.3.0.0/24 accessible via 193.1.0.2 (eth0, Router 2)

  OR (since all remote IP networks are routed through the same next-hop router)

    - Using default route: 0.0.0.0/0 via through 193.1.0.2 (eth0, Router 2)

IP Static Routing

- Static routing does not scale well when network grows:
  - Administrative burden to maintain routes may become excessive.
- Static routing does not react to network topology changes:
  - If a link in the path of a static route fails, connectivity is lost even if there are alternative routing paths.
  - The link failure must be administratively detected and solved (i.e., either solving the failure or configuring a new static route), causing the connectivity lost for a long time.
- Nevertheless, static routing is useful:
  - when the administrator needs total control over all routing paths, for reasons like security of QoS (Quality of Service)
  - when a backup to a dynamically learned route is necessary (if the dynamic routing protocol fails)
  - when there is a single routing path towards an IP network
    - example: a home router that connects to its Internet Service Provider (ISP) only needs to have a default route toward all outside IP networks

IP Dynamic Routing

- Dynamic routing is implemented through protocols running in the routers.
- Dynamic routing protocols, as their name suggests, are used to dynamically exchange routing information between routers.
- Their implementation allows network routing to dynamically adjust to changing network conditions, and to ensure that efficient and redundant routing continues in spite of any changes:
  - When the network topology changes (a failure is detected as a topology change), the new information is dynamically exchanged throughout the network, and each router updates its routing table to reflect the changes.
- Routers exchange routing information only with other routers running the same routing protocol.

Longest Match Routing

- **Longest Match Routing**: when an incoming IP packet in a router has more than one matching entry in its IP routing table, the routing decision uses the entry with the longest prefix

- Example: The IP routing table of a router includes the routes:

  a) 192.168.1.0/24 via ...

  b) 192.168.0.0/16 via ...

  c) 0.0.0.0/0 via ...

- An incoming IP packet to 192.168.1.12 matches all entries:
  - router forwards the packet based on entry: **`192.168.1.0/24 via ...`**
- An incoming IP packet to 192.168.8.14 matches entries b) and c):
  - router forwards the packet based on entry: **`192.168.0.0/16 via ...`**
- An incoming IP packet to 192.1.8.14 matches only entry c):
  - router forwards the packet based on entry: **`0.0.0.0/0 via ...`**

Administrative Distance

- Routers use an **Administrative Distance** value to select the routing paths when they learn the same remote IP networks with different routing methods.
  - The method <u>with the lowest Administrative Distance is preferred</u>
- The Administrative Distance has default values but can be configured differently by the administrator.

- Example with default values:
  - Static [**1**/1] 192.168.1.0/24 via … ← Chosen!
  - RIP [**120**/2] 192.168.1.0/24 via …
  - OSPF [**110**/5] 192.168.1.0/24 via ...

- If Static Routing is used as backup to the dynamic routing protocols, its Administrative Distance must be configured with a value larger than the value(s) of the protocol(s) in use

Equal Cost Multi-Pathing (ECMP)

- **Equal Cost Multi-Pathing**: when the IP routing table of a router has multiple next-hop routers to the same remote IP network, the router applies load balancing:
  - the router splits the packets equally by the different next-hop routers towards the remote IP network

Example:

```
R   192.168.30.0/24 [120/1] via 192.168.20.2, 00:00:08, FastEthernet0/1
                    [120/1] via 192.168.10.1, 00:00:09, FastEthernet0/0
```

Router splits the IP packets towards the remote IP network 192.168.30.0/24 in equal percentage via next-hop router addresses 192.168.20.1 and 192.168.10.1

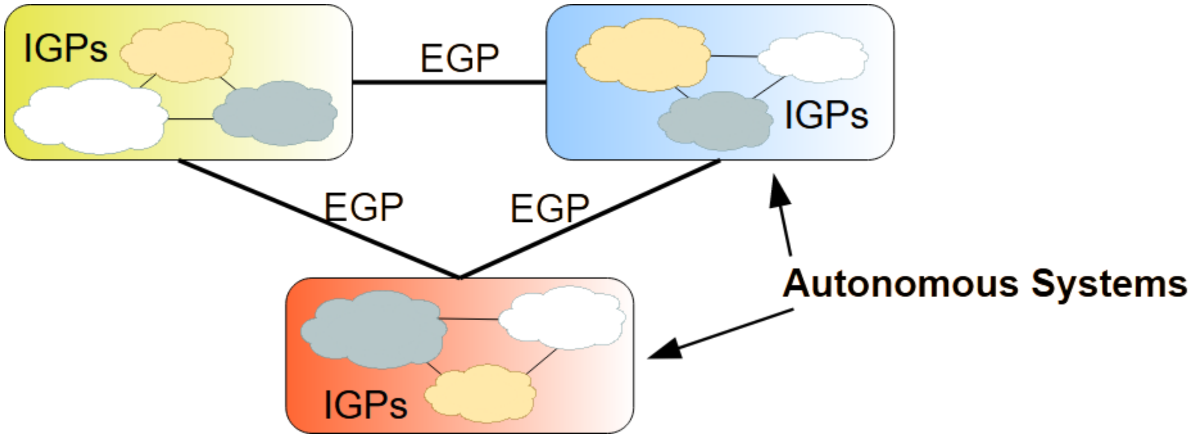
- The equally split of the packets by the different next-hop routers can be packet-based of flow based (next slide)

Load Balancing in ECMP

- **Packet-based load balancing** (the old approach):
  - IP packets are split with a round-robin algorithm
  - Introduces high jitter (and possibly out-of-order reception) in IP flows when the delay of routing paths is significantly different between them
  - High jitter and out-of-order reception of IP packets penalises the performance of the TCP congestion-control algorithm
- **Flow-based load balancing** (the current approach):
  - IP flows are dynamically assigned among all next-hop routers with the same probability (through hashing)
    - An IP flow is defined by a combination of IP source and destination addresses, protocol (UDP or TCP), source and destination port numbers
  - IP packets are forwarded based on the next-hop router assigned to the flow they belong:
    - IP packets of the same flow are all routed through the same path (eliminating the high jitter and out-of-order reception)
    - it requires hashing, whose complexity is negligible to the processing capacity of current routers

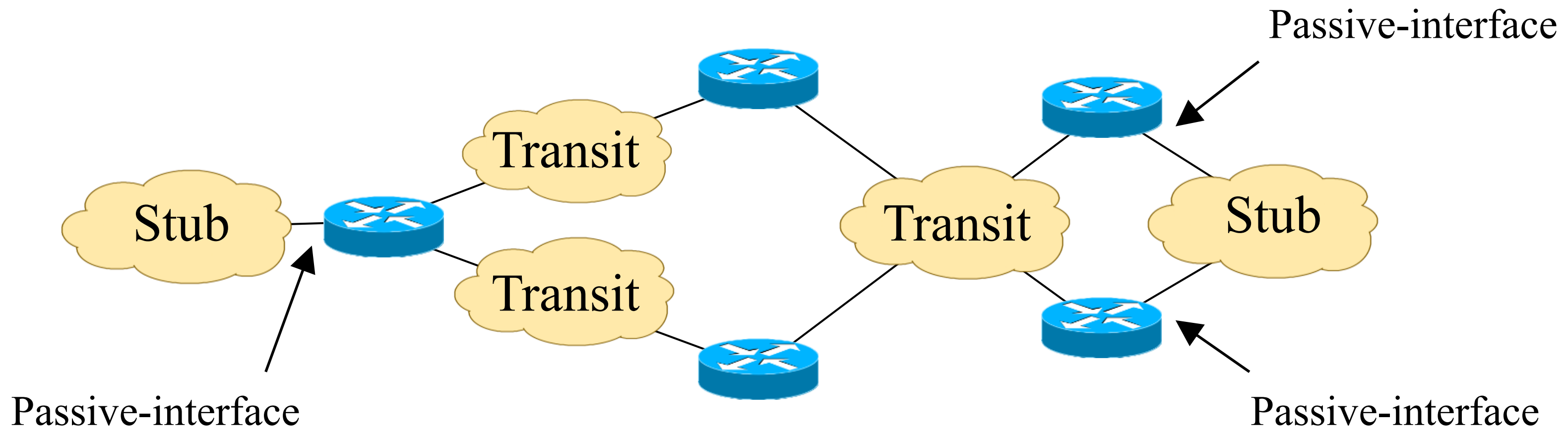
Internet Autonomous Systems

- **AS (Autonomous System)** – set of routers/networks with a common routing policy and under the same administration.
- Routing inside an AS is performed by **IGPs (Interior Gateway Protocols)** such as RIP, OSPF, IS-IS and EIGRP.
- Routing between AS is performed by **EGPs (Exterior Gateway Protocols)** such as BGP (which has become the standard de facto).
- IGPs and EGPs have different objectives:
  - IGPs: optimize routing performance
  - EGPs: optimize routing performance but highly constrained by economic, political and security policies.



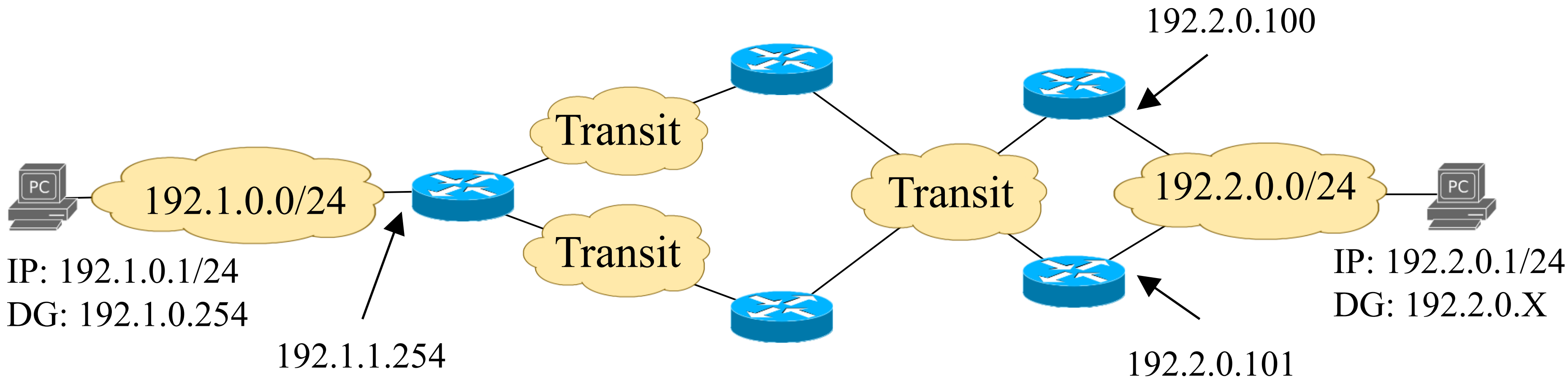
Types of Networks between Routers inside an AS

- Transit Networks:
  - used by routers to route IP packets between other IP networks
  - routing protocols use them to exchange routing information
- Stub Networks:
  - networks with one attached router
  - networks with more than one attached routers, but not used to route IP packets between other IP networks
  - in both cases, the router interfaces should be set as passive-interfaces so that routing protocols do not exchange routing information through them



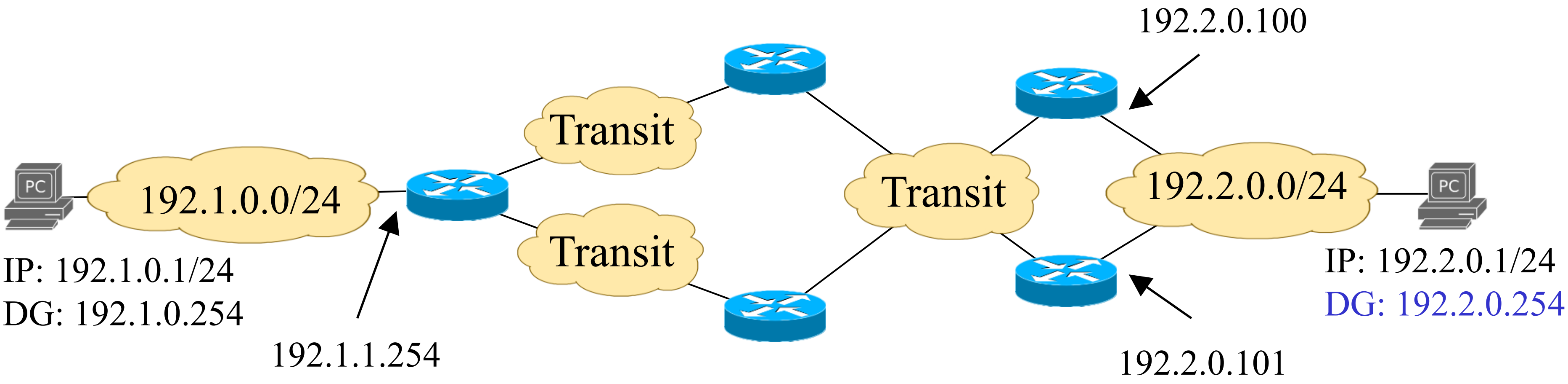
Virtual Router Redundancy Protocol (VRRP)

- When an IP network is connected to a single router, the IP address of the router must be the Default Gateway (DG) to all hosts of the network
  - In the figure below, the hosts in the network 192.1.0.0/24 must be configured with the Default Gateway 192.1.1.254
- When an IP network is connected to multiple routers, the DG address (configured in the hosts) must represent one of the routers
  - In the figure below, the hosts in the network 192.2.0.0/24 can be configured with the DG 192.2.0.100 or 192.2.0.101
  - This solution is not robust to failures: if the configured DG fails, the IP connectivity fails although there is an alternative router that can act as DG



Virtual Router Redundancy Protocol (VRRP)

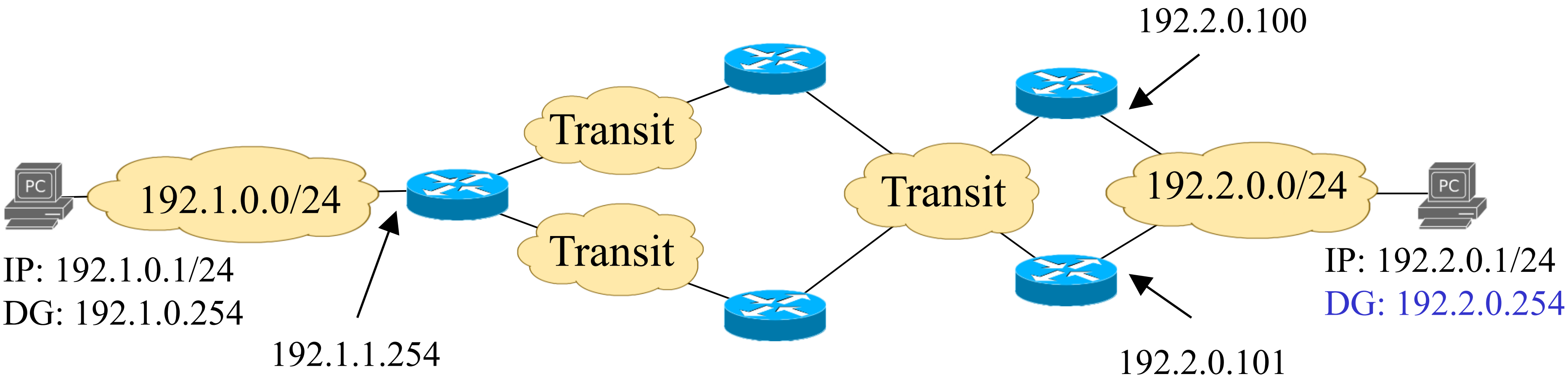
- Although it is possible to configure alternative DGs in hosts, this solution is not ideal:
  - All DG addresses must be configured in hosts (through DHCP)
  - It takes time for a host to assume that the primary DG has failed before it starts using the alternative DG, leading to significant connectivity failure times
- VRRP is a standard protocol defined by the IETF to protect the network against a failure of the DG when there are multiple DG alternatives.
- With VRRP, a cluster of routers is configured to act as a single router to a LAN with a virtual IP address (to be used as the DG in the hosts).
  - In the example below, the configured virtual DG is 192.2.0.254



Virtual Router Redundancy Protocol (VRRP)

VRRP works as follows:

- All routers of the cluster send VRRP announcement messages to a standard defined multicast address:
  - IPv4: 224.0.0.18
  - IPv6: FF02::12
- The “best router” (the one with the highest VRRP priority or the highest IP address, if the priority is the same) acts as master and keeps sending the VRRP messages (default advertisement interval = 1 second)
- If the master fails (i.e., if it does not send VRRP messages for 3 seconds), the “best router” among the remaining ones becomes the master.



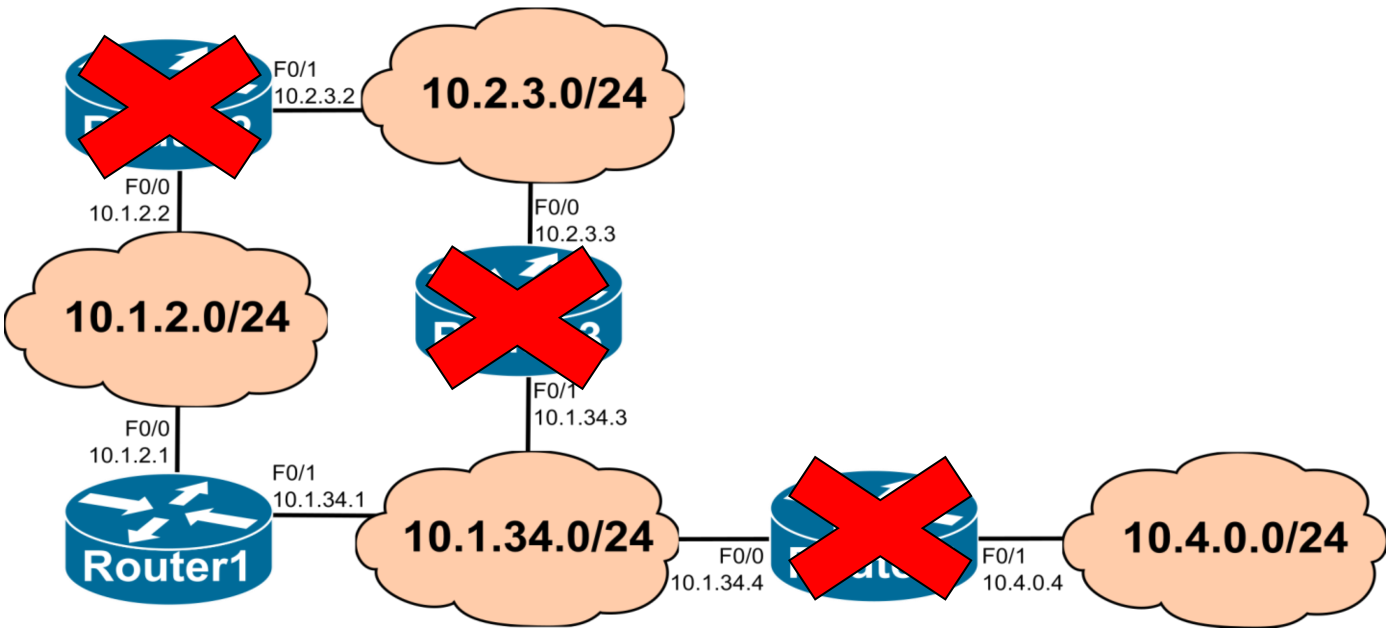
Types of IGP Dynamic Routing Protocols

- Distance vector
  - Each router builds and maintains a vector (i.e., a list) of known IP networks and associated distances (costs), based on the information sent periodically between neighbour routers.
  - Each router determines the shortest paths to all remote IP networks based on the distributed and asynchronous Bellman-Ford algorithm.
  - Examples: RIPv2, IGRP, EIGRP.
- Link state
  - Each router learns and maintains a database with the complete network topology view and uses a centralized shortest path algorithm to determine the routing paths from it to all remote IP networks.
  - The information required to built the network topology database on each router is obtained by a flooding process.
  - Network information is only exchanged on bootstrap and after any topology change.
  - Examples: OSPF, IS-IS.

RIP (Routing Information Protocol)

- RIP is a *distance vector* routing protocol
  - Each router maintains a list of known IP networks and, for each network, an estimation of the cost to reach it – this is called a distance vector.
  - Each router periodically sends to its neighbour routers its own distance vector (partially or complete).
  - Each router uses the distance vector received from its neighbours to update its own distance vector.
- The path cost from a router to a remote IP network destination is given by the number of intermediate routers along the path.
  - Maximum cost is 15.
  - A cost of 16 is considered infinite (i.e., an unreachable destination).
- Each router determines the entries of its IP routing table based on its current distance vector:
  - For each remote IP network in the distance vector, the router adds an entry towards each neighbour router providing the lowest cost.

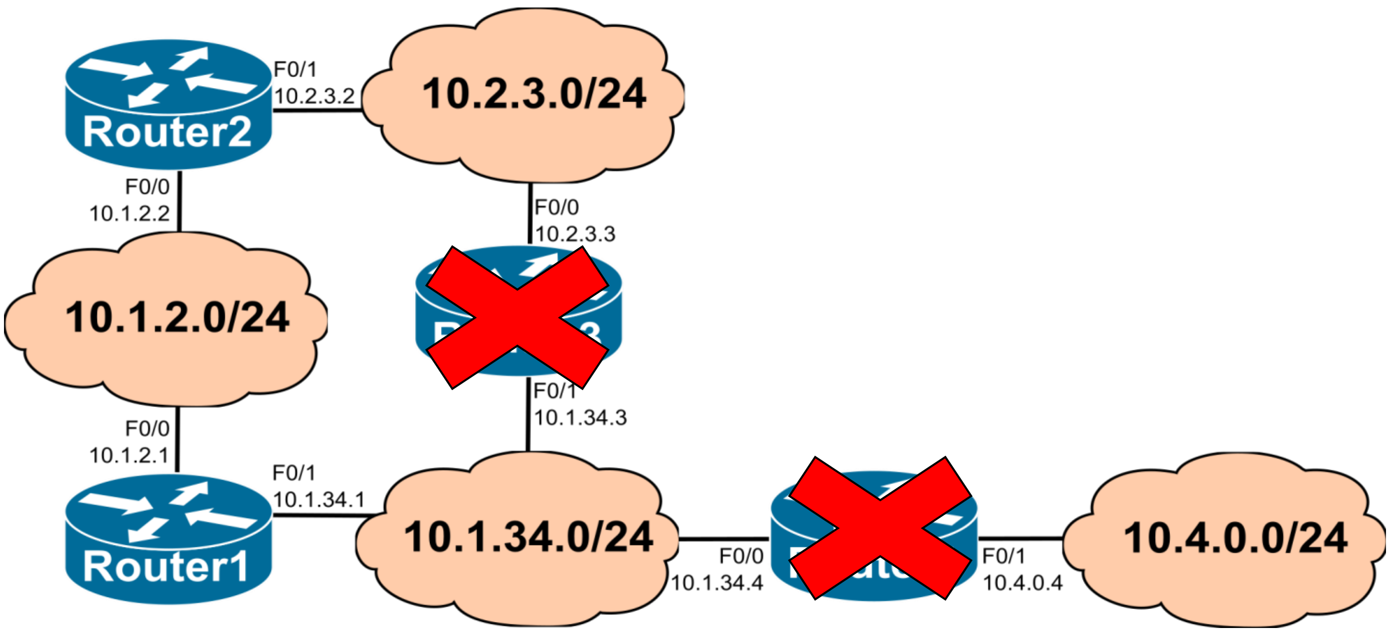
Routing Tables with RIP
Router1 starts




```
Router1
C 10.1.2.0/24 is directly connected, FastEthernet0/0
C 10.1.34.0/24 is directly connected, FastEthernet0/1
```

Routing Tables with RIP
Router2 starts

```
Router2
C 10.1.2.0/24 is directly connected, FastEthernet0/0
R 10.1.34.0/24 [120/1] via 10.1.2.1, 00:00:11, FastEthernet0/0
C 10.2.3.0/24 is directly connected, FastEthernet0/1
```

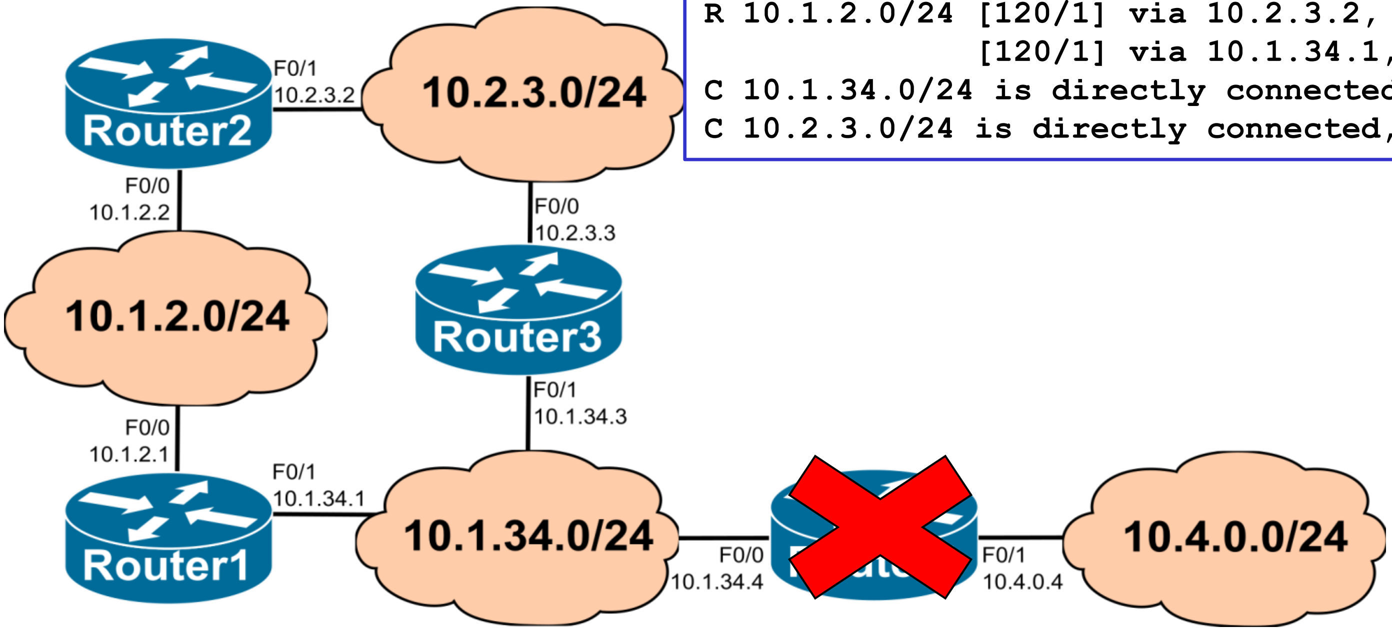



```
Router1
C 10.1.2.0/24 is directly connected, FastEthernet0/0
C 10.1.34.0/24 is directly connected, FastEthernet0/1
R 10.2.3.0/24 [120/1] via 10.1.2.2, 00:00:05, FastEthernet0/0
```

Routing Tables with RIP

Router3 starts

```
Router2
C 10.1.2.0/24 is directly connected, FastEthernet0/0
R 10.1.34.0/24 [120/1] via 10.2.3.3, 00:00:06, FastEthernet0/1
               [120/1] via 10.1.2.1, 00:00:13, FastEthernet0/0
C 10.2.3.0/24 is directly connected, FastEthernet0/1
```



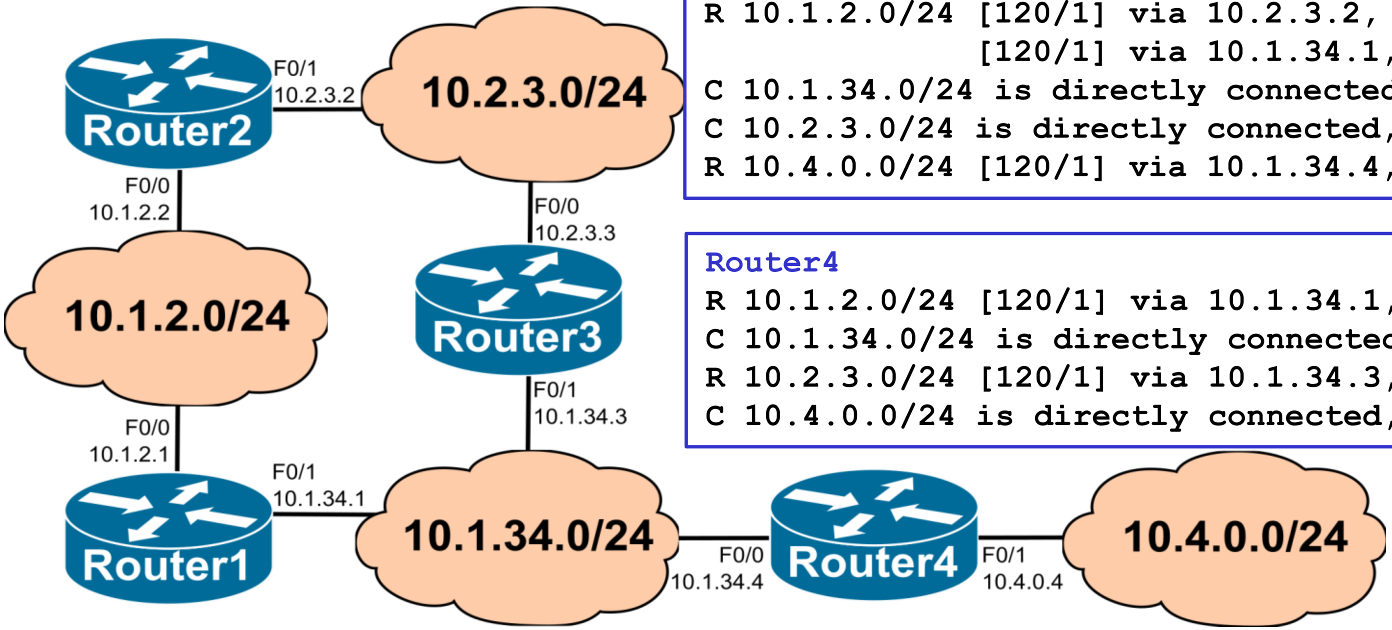

```
Router3
R 10.1.2.0/24 [120/1] via 10.2.3.2, 00:00:08, FastEthernet0/0
              [120/1] via 10.1.34.1, 00:00:18, FastEthernet0/1
C 10.1.34.0/24 is directly connected, FastEthernet0/1
C 10.2.3.0/24 is directly connected, FastEthernet0/0
```

```
Router1
C 10.1.2.0/24 is directly connected, FastEthernet0/0
C 10.1.34.0/24 is directly connected, FastEthernet0/1
R 10.2.3.0/24 [120/1] via 10.1.34.3, 00:00:04, FastEthernet0/1
              [120/1] via 10.1.2.2, 00:00:12, FastEthernet0/0
```

Routing Tables with RIP

Router4 starts

```
Router2
C 10.1.2.0/24 is directly connected, FastEthernet0/0
R 10.1.34.0/24 [120/1] via 10.2.3.3, 00:00:14, FastEthernet0/1
               [120/1] via 10.1.2.1, 00:00:07, FastEthernet0/0
C 10.2.3.0/24 is directly connected, FastEthernet0/1
R 10.4.0.0/24 [120/2] via 10.2.3.3, 00:00:14, FastEthernet0/1
              [120/2] via 10.1.2.1, 00:00:07, FastEthernet0/0
```



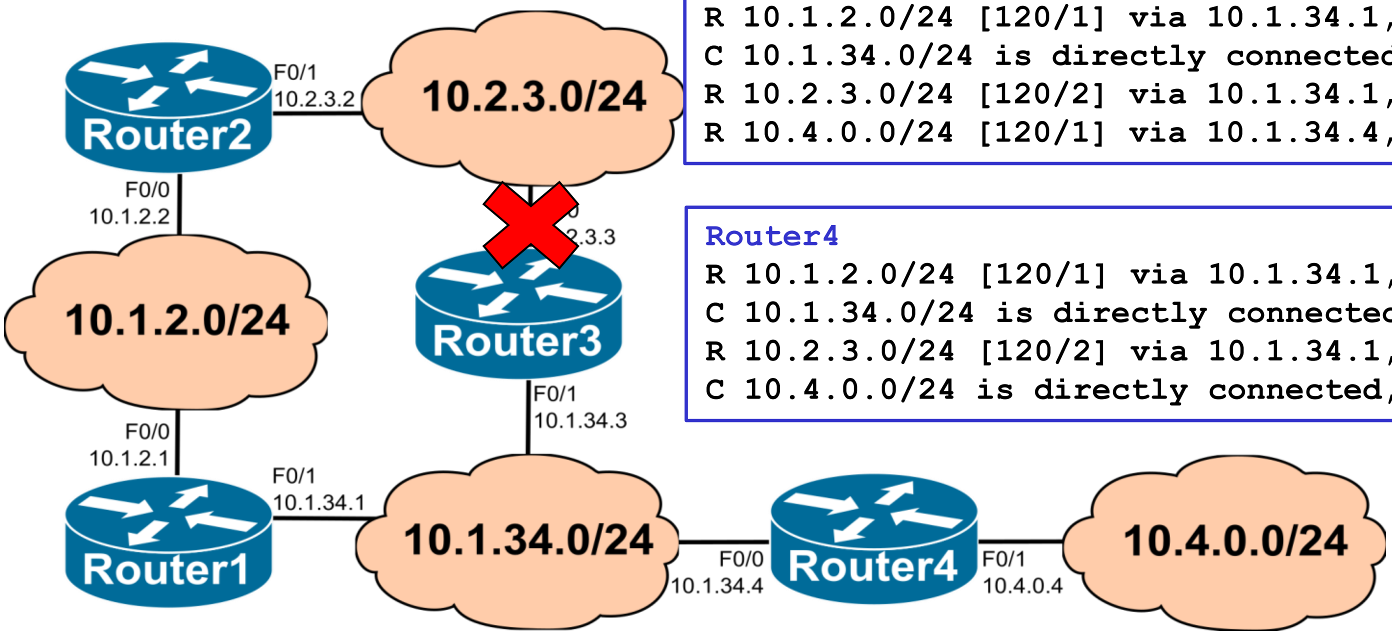

```
Router3
R 10.1.2.0/24 [120/1] via 10.2.3.2, 00:00:09, FastEthernet0/0
              [120/1] via 10.1.34.1, 00:00:11, FastEthernet0/1
C 10.1.34.0/24 is directly connected, FastEthernet0/1
C 10.2.3.0/24 is directly connected, FastEthernet0/0
R 10.4.0.0/24 [120/1] via 10.1.34.4, 00:00:14, FastEthernet0/1
```

```
Router4
R 10.1.2.0/24 [120/1] via 10.1.34.1, 00:00:15, FastEthernet0/0
C 10.1.34.0/24 is directly connected, FastEthernet0/0
R 10.2.3.0/24 [120/1] via 10.1.34.3, 00:00:12, FastEthernet0/0
C 10.4.0.0/24 is directly connected, FastEthernet0/1
```

```
Router1
C 10.1.2.0/24 is directly connected, FastEthernet0/0
C 10.1.34.0/24 is directly connected, FastEthernet0/1
R 10.2.3.0/24 [120/1] via 10.1.34.3, 00:00:20, FastEthernet0/1
              [120/1] via 10.1.2.2, 00:00:04, FastEthernet0/0
R 10.4.0.0/24 [120/1] via 10.1.34.4, 00:00:12, FastEthernet0/1
```

Routing Tables with RIP (after an interface failure)

```
Router2 (AFTER 3 minutes TIMEOUT)
C 10.1.2.0/24 is directly connected, FastEthernet0/0
R 10.1.34.0/24 [120/1] via 10.1.2.1, 00:00:25, FastEthernet0/0
C 10.2.3.0/24 is directly connected, FastEthernet0/1
R 10.4.0.0/24 [120/2] via 10.1.2.1, 00:00:25, FastEthernet0/0
```




```
Router3
R 10.1.2.0/24 [120/1] via 10.1.34.1, 00:00:22, FastEthernet0/1
C 10.1.34.0/24 is directly connected, FastEthernet0/1
R 10.2.3.0/24 [120/2] via 10.1.34.1, 00:00:22, FastEthernet0/1
R 10.4.0.0/24 [120/1] via 10.1.34.4, 00:00:19, FastEthernet0/1
```

```
Router4
R 10.1.2.0/24 [120/1] via 10.1.34.1, 00:00:18, FastEthernet0/0
C 10.1.34.0/24 is directly connected, FastEthernet0/0
R 10.2.3.0/24 [120/2] via 10.1.34.1, 00:00:18, FastEthernet0/0
C 10.4.0.0/24 is directly connected, FastEthernet0/1
```

```
Router1
C 10.1.2.0/24 is directly connected, FastEthernet0/0
C 10.1.34.0/24 is directly connected, FastEthernet0/1
R 10.2.3.0/24 [120/1] via 10.1.2.2, 00:00:01, FastEthernet0/0
R 10.4.0.0/24 [120/1] via 10.1.34.4, 00:00:24, FastEthernet0/1
```

RIP Messages

- RIP Response message
  - It contains the distance vector of the sender router. It is sent:
    1. Periodically (~30 seconds by default).
    2. When the distance vector information changes (***triggered updates***).
    3. In response to a RIP Request from a neighbour router.
  - In RIP version 2, the RIP Response messages are sent to:
    - the standard defined multicast address 224.0.0.9 (in cases 1 and 2)
    - the unicast address of the router that sent the RIP Request (in case 3)
- RIP Request message (Optional)
  - Sent by a router when it starts (bootstrap) or, when the validity of some of the distance vector information has expired (default timeout = 180 seconds)
  - It may request specific information (a specific network), or the complete distance vector of the neighbour router.
- Both RIP messages are sent through UDP (port number = 520)

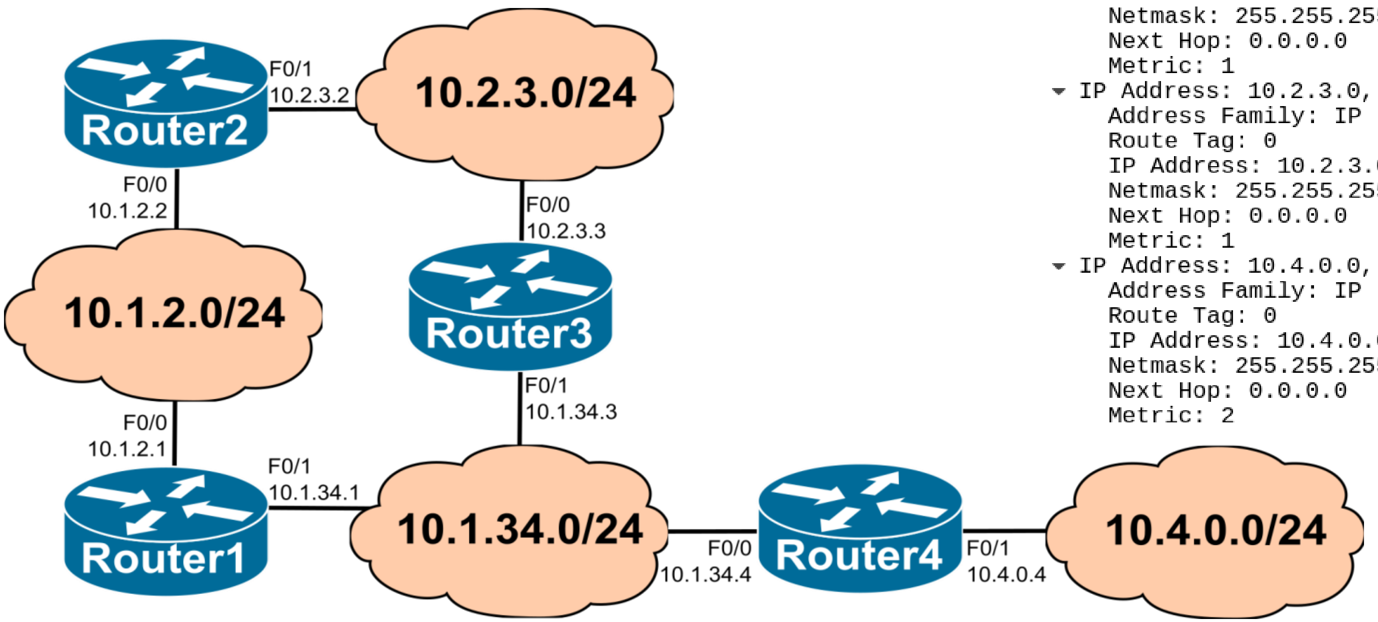
RIPv2 Response messages <u>without split-horizon</u>

- ***The RIP Response messages are sent by each router with its complete distance vector through all interfaces***

- The metric announced to each IP network is the number of hops that the neighbour router will have if the neighbour uses it as the next hop to the network

Sent by Router 3 through F0/0

```
▸ Internet Protocol Version 4, Src: 10.2.3.3, Dst: 224.0.0.9
▸ User Datagram Protocol, Src Port: 520, Dst Port: 520
▾ Routing Information Protocol
    Command: Response (2)
    Version: RIPv2 (2)
  ▾ IP Address: 10.1.2.0, Metric: 2
      Address Family: IP (2)
      Route Tag: 0
      IP Address: 10.1.2.0
      Netmask: 255.255.255.0
      Next Hop: 10.2.3.2
      Metric: 2
  ▾ IP Address: 10.1.34.0, Metric: 1
      Address Family: IP (2)
      Route Tag: 0
      IP Address: 10.1.34.0
      Netmask: 255.255.255.0
      Next Hop: 0.0.0.0
      Metric: 1
  ▾ IP Address: 10.2.3.0, Metric: 1
      Address Family: IP (2)
      Route Tag: 0
      IP Address: 10.2.3.0
      Netmask: 255.255.255.0
      Next Hop: 0.0.0.0
      Metric: 1
  ▾ IP Address: 10.4.0.0, Metric: 2
      Address Family: IP (2)
      Route Tag: 0
      IP Address: 10.4.0.0
      Netmask: 255.255.255.0
      Next Hop: 0.0.0.0
      Metric: 2
```

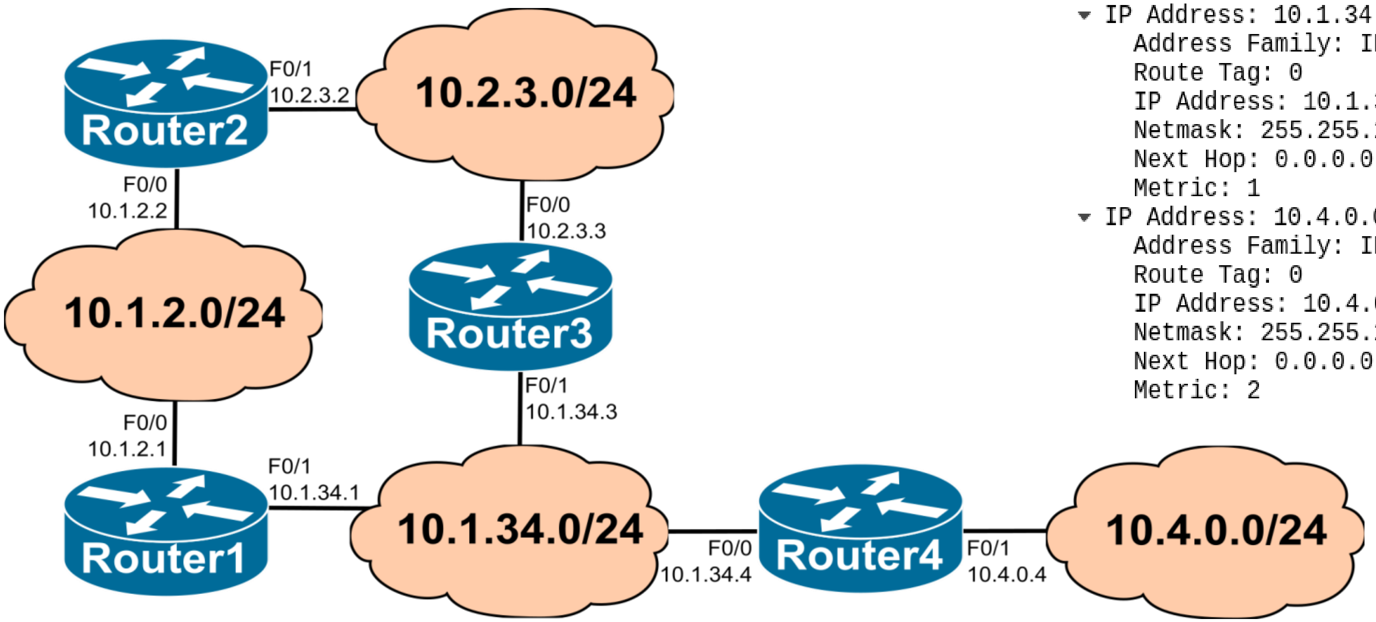


RIPv2 Response messages <u>with split-horizon</u>

- ***The RIP Response messages sent by a router through an interface include only the IP networks whose shortest path is not through the interface.***
  - The split-horizon shortens the average number of iterations required for the convergence of the IP routing tables when the network topology changes
- In the example, Router 3 does not announce 10.1.2.0/24 through F0/0 because one of the two shortest paths from Router 3 to 10.1.2.0/24 is through its F0/0 interface.

Sent by Router 3 through F0/0

```
▸ Internet Protocol Version 4, Src: 10.2.3.3, Dst: 224.0.0.9
▸ User Datagram Protocol, Src Port: 520, Dst Port: 520
▾ Routing Information Protocol
    Command: Response (2)
    Version: RIPv2 (2)
  ▾ IP Address: 10.1.34.0, Metric: 1
      Address Family: IP (2)
      Route Tag: 0
      IP Address: 10.1.34.0
      Netmask: 255.255.255.0
      Next Hop: 0.0.0.0
      Metric: 1
  ▾ IP Address: 10.4.0.0, Metric: 2
      Address Family: IP (2)
      Route Tag: 0
      IP Address: 10.4.0.0
      Netmask: 255.255.255.0
      Next Hop: 0.0.0.0
      Metric: 2
```



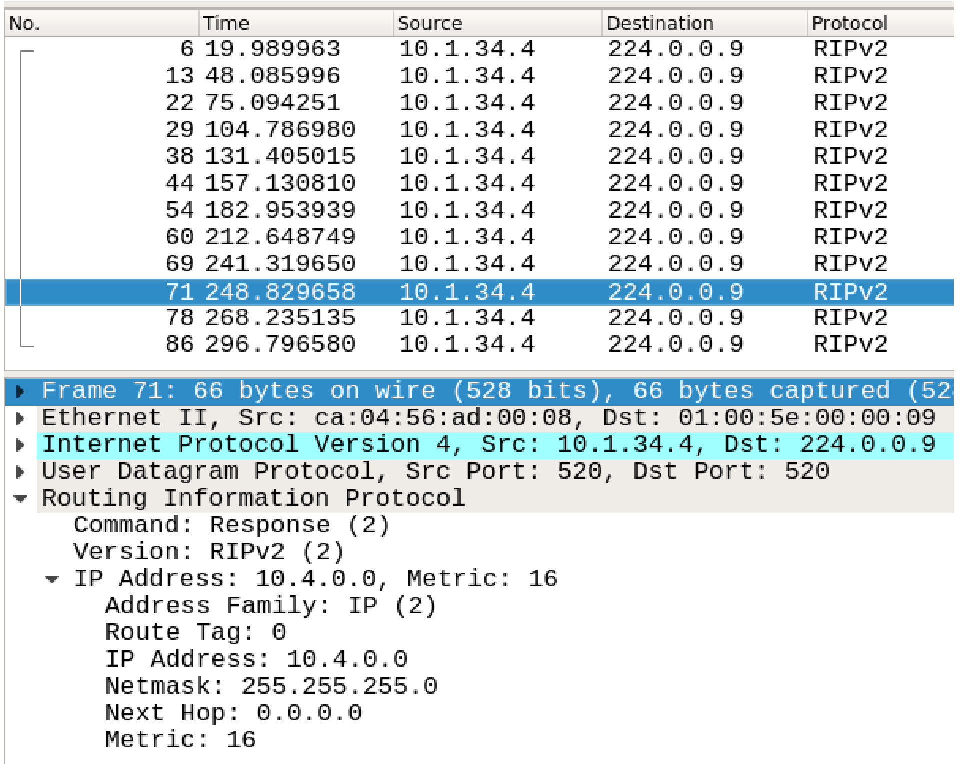
Triggered Updates

- RIP Response messages are sent periodically to continuously check the connectivity between neighbour routers
- ***Triggered updates*** are RIP Response messages sent by a router to its neighbour routers immediately after a change of its current distance vector (instead of waiting for the time instant of the next periodic RIP Response):
  - if the cost of a known IP network changes,
  - if a new IP network is added,
  - if a previously known IP network is removed.
- In this way, neighbour routers update their distance vector faster and overall convergence becomes also faster.

Triggered Updates (Illustration)

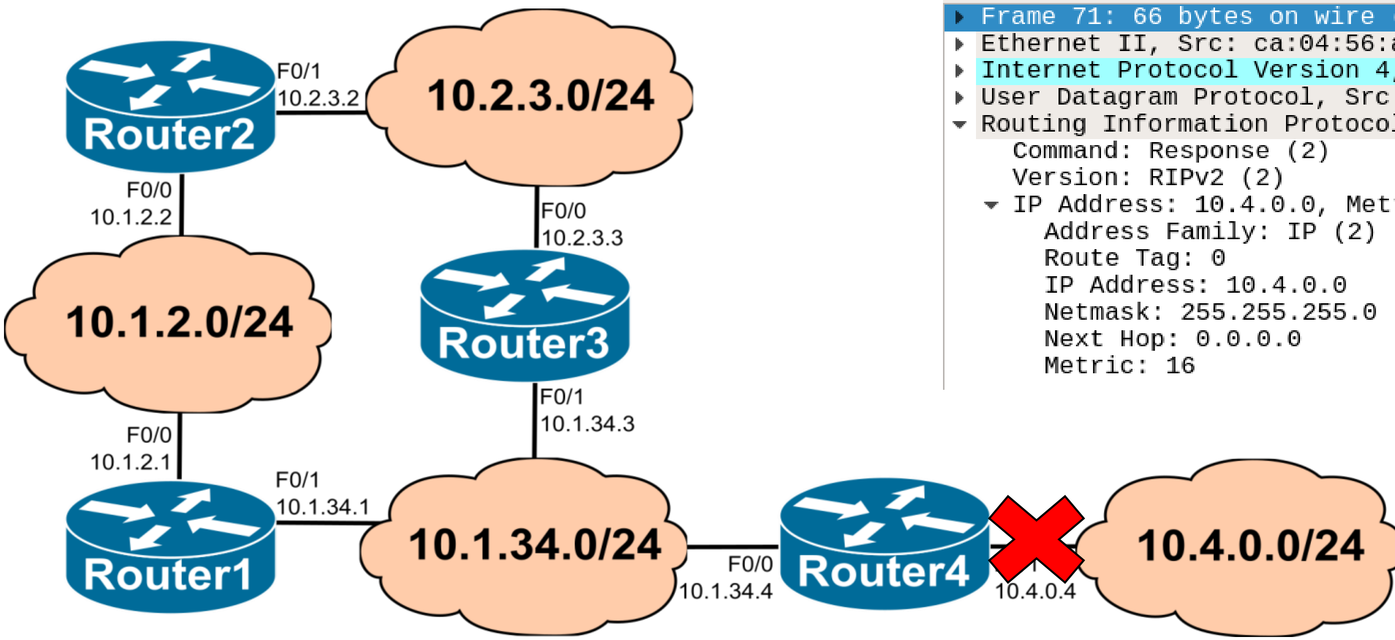
- The interface F0/1 of Router 4 fails, causing Router 4 to lose connectivity with network 10.4.0.0/24
- Router 4 immediately sends through F0/0 a triggered update announcing the network 10.4.0.0/24 with cost 16 (which means not reachable)
- In this case, the network 10.4.0.0/24 will be quickly removed from the IP routing table of all other routers

Sent by Router 4 through F0/0



No.	Time	Source	Destination	Protocol
6	19.989963	10.1.34.4	224.0.0.9	RIPv2
13	48.085996	10.1.34.4	224.0.0.9	RIPv2
22	75.094251	10.1.34.4	224.0.0.9	RIPv2
29	104.786980	10.1.34.4	224.0.0.9	RIPv2
38	131.405015	10.1.34.4	224.0.0.9	RIPv2
44	157.130810	10.1.34.4	224.0.0.9	RIPv2
54	182.953939	10.1.34.4	224.0.0.9	RIPv2
60	212.648749	10.1.34.4	224.0.0.9	RIPv2
69	241.319650	10.1.34.4	224.0.0.9	RIPv2
71	248.829658	10.1.34.4	224.0.0.9	RIPv2
78	268.235135	10.1.34.4	224.0.0.9	RIPv2
86	296.796580	10.1.34.4	224.0.0.9	RIPv2

```
Frame 71: 66 bytes on wire (528 bits), 66 bytes captured (52
Ethernet II, Src: ca:04:56:ad:00:08, Dst: 01:00:5e:00:00:09
Internet Protocol Version 4, Src: 10.1.34.4, Dst: 224.0.0.9
User Datagram Protocol, Src Port: 520, Dst Port: 520
Routing Information Protocol
  Command: Response (2)
  Version: RIPv2 (2)
  IP Address: 10.4.0.0, Metric: 16
    Address Family: IP (2)
    Route Tag: 0
    IP Address: 10.4.0.0
    Netmask: 255.255.255.0
    Next Hop: 0.0.0.0
    Metric: 16
```

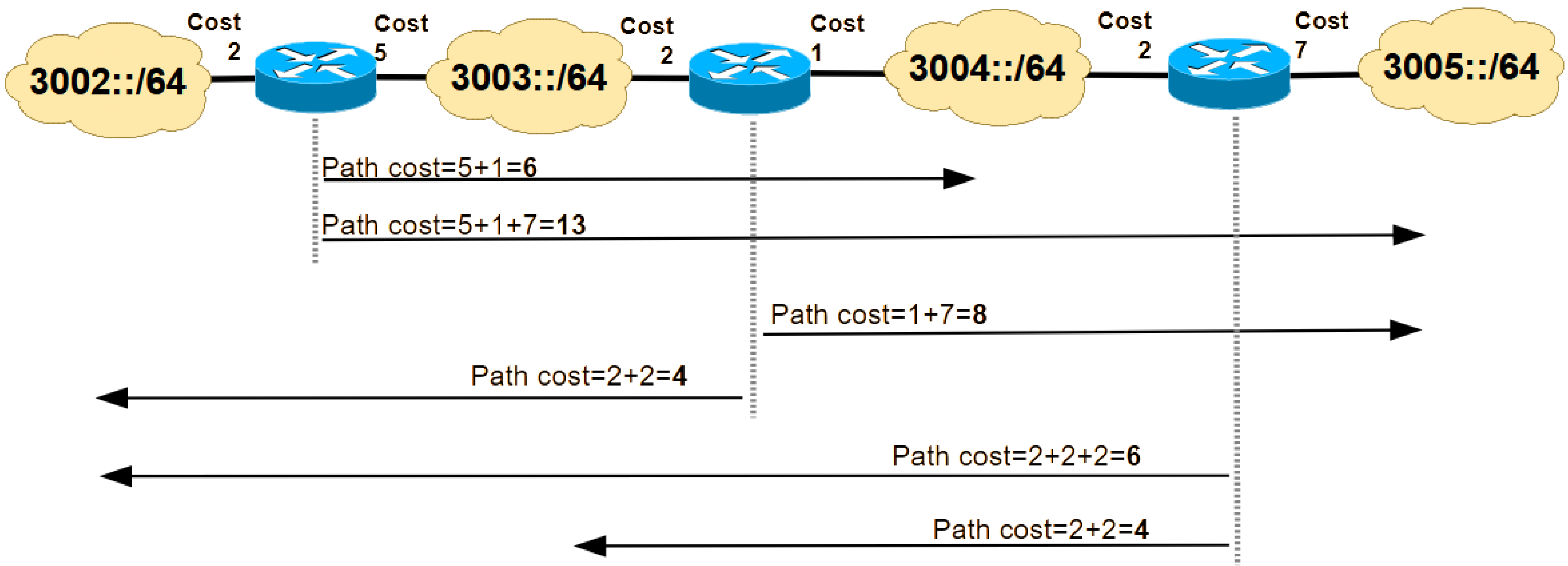


RIP Next Generation (RIPng) for IPv6 Routing

- RIPng (for IPv6) is similar to RIP version 2 (RIPv2):
  - Distance-vector concept, maximum cost is 15 (16 is infinite), split-horizon, triggered updates, messages transported over UDP
- Differences from RIPv2 to RIPng:
  - RIPng messages are sent through link-local IPv6 addresses (FE80::/64).
    - Uses multicast group address FF02::9 as the destination address for RIP Response messages.
  - A cost is assigned to each output interface.
  - The cost of a routing path is the sum of the costs of the output interfaces towards the destination (illustration in the next slide):
    - allows the operator to control the resulting routing paths.
  - In the split-horizon configuration, routers also include the networks connected to the sending interface:
    - since messages are sent through link-local IPv6 addresses, each router does not know if the neighbour routers on a given interface are also connected to the same global IPv6 networks.

RIPng Path Costs

- Each router interface has an associated RIPng cost (default value = 1).
- The total cost of a routing path from a router to a remote IPv6 network is the sum of the costs of the output interfaces of each router along the path.
- With the infinity metric value at 16, the costs must be carefully configured to prevent any routing path to cost more than 15.



RIPng Response Messages with Split-Horizon

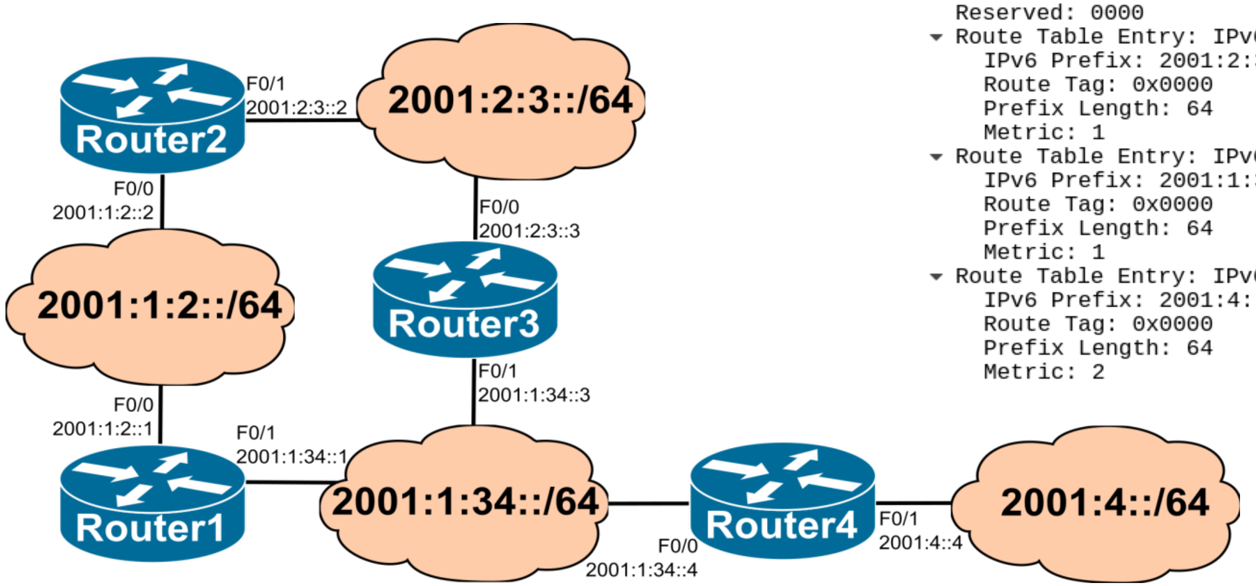
Sent by Router 2 through F0/1

```
▸ Internet Protocol Version 6, Src: fe80::c802:54ff:fef5:6, Dst: ff02::9
▸ User Datagram Protocol, Src Port: 521, Dst Port: 521
▾ RIPng
    Command: Response (2)
    Version: 1
    Reserved: 0000
  ▾ Route Table Entry: IPv6 Prefix: 2001:1:2::/64 Metric: 1
      IPv6 Prefix: 2001:1:2::
      Route Tag: 0x0000
      Prefix Length: 64
      Metric: 1
  ▾ Route Table Entry: IPv6 Prefix: 2001:2:3::/64 Metric: 1
      IPv6 Prefix: 2001:2:3::
      Route Tag: 0x0000
      Prefix Length: 64
      Metric: 1
```

The IPv6 global network 2001:2:3::/64 is included in the RIPng Response messages exchanged through this network

Sent by Router 3 through F0/0

```
▸ Internet Protocol Version 6, Src: fe80::c803:56ff:fe0a:8, Dst: ff02::9
▸ User Datagram Protocol, Src Port: 521, Dst Port: 521
▾ RIPng
    Command: Response (2)
    Version: 1
    Reserved: 0000
  ▾ Route Table Entry: IPv6 Prefix: 2001:2:3::/64 Metric: 1
      IPv6 Prefix: 2001:2:3::
      Route Tag: 0x0000
      Prefix Length: 64
      Metric: 1
  ▾ Route Table Entry: IPv6 Prefix: 2001:1:34::/64 Metric: 1
      IPv6 Prefix: 2001:1:34::
      Route Tag: 0x0000
      Prefix Length: 64
      Metric: 1
  ▾ Route Table Entry: IPv6 Prefix: 2001:4::/64 Metric: 2
      IPv6 Prefix: 2001:4::
      Route Tag: 0x0000
      Prefix Length: 64
      Metric: 2
```

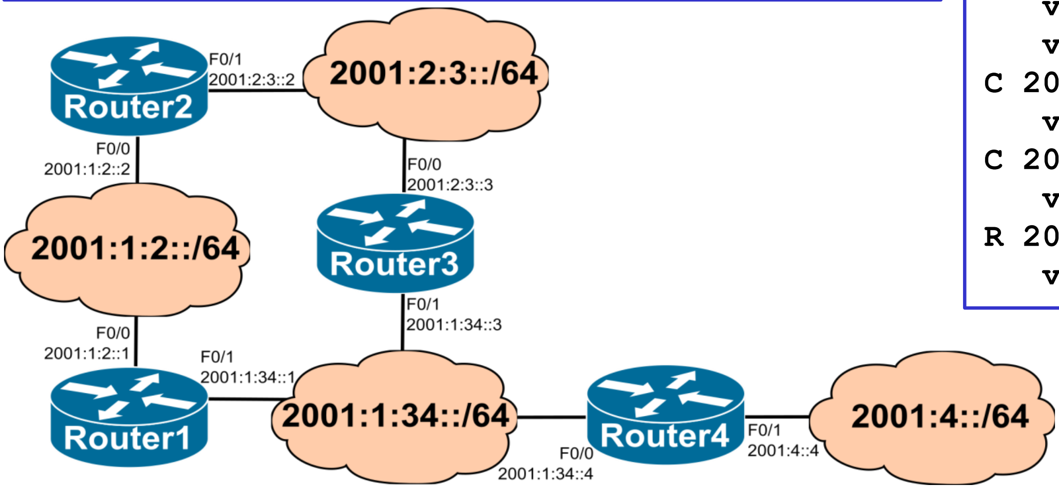


IPv6 Routing Tables with RIPng

Assuming the default cost values

```
Router2
C 2001:1:2::/64 [0/0]
   via FastEthernet0/0, directly connected
R 2001:1:34::/64 [120/2]
   via FE80::C801:54FF:FE41:8, FastEthernet0/0
   via FE80::C803:56FF:FE0A:8, FastEthernet0/1
C 2001:2:3::/64 [0/0]
   via FastEthernet0/1, directly connected
R 2001:4::/64 [120/3]
   via FE80::C801:54FF:FE41:8, FastEthernet0/0
   via FE80::C803:56FF:FE0A:8, FastEthernet0/
```

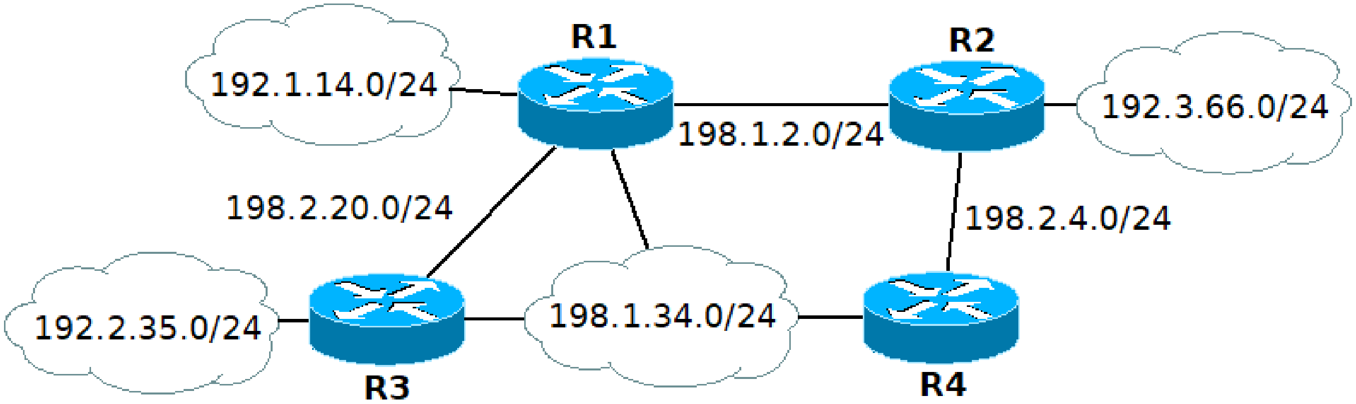
```
Router3
R 2001:1:2::/64 [120/2]
   via FE80::C802:54FF:FEF5:6, FastEthernet0/0
   via FE80::C801:54FF:FE41:6, FastEthernet0/1
C 2001:1:34::/64 [0/0]
   via FastEthernet0/1, directly connected
C 2001:2:3::/64 [0/0]
   via FastEthernet0/0, directly connected
R 2001:4::/64 [120/2]
   via FE80::C804:56FF:FEAD:8, FastEthernet0/1
```




```
Router1
C 2001:1:2::/64 [0/0]
   via FastEthernet0/0, directly connected
C 2001:1:34::/64 [0/0]
   via FastEthernet0/1, directly connected
R 2001:2:3::/64 [120/2]
   via FE80::C802:54FF:FEF5:8, FastEthernet0/0
   via FE80::C803:56FF:FE0A:6, FastEthernet0/1
R 2001:4::/64 [120/2]
   via FE80::C804:56FF:FEAD:8, FastEthernet0/1
```

```
Router4
R 2001:1:2::/64 [120/2]
   via FE80::C801:54FF:FE41:6, FastEthernet0/0
C 2001:1:34::/64 [0/0]
   via FastEthernet0/0, directly connected
R 2001:2:3::/64 [120/2]
   via FE80::C803:56FF:FE0A:6, FastEthernet0/0
C 2001:4::/64 [0/0]
   via FastEthernet0/1, directly connected
```

Example 1 (RIP)



Consider the IP network of the figure. The host part of each IP address is the number of the router's name. All routers are configured with RIPv2 with split-horizon in all IP networks.

a) What is the IP routing table of R3?

b) What is the distance vector announced by R4 through its interface to network 198.1.34.0/24?

c) How should R2 be configured so that it does not send RIP messages through its interface of network 192.3.66.0/24?

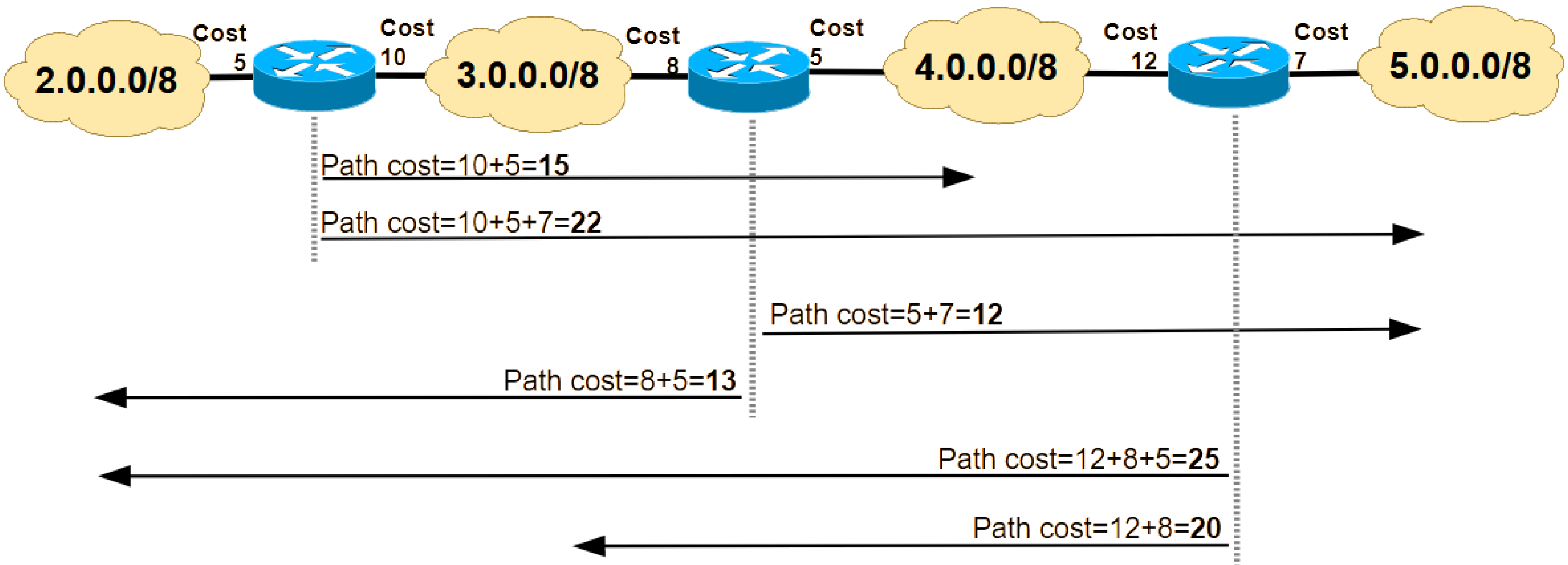
d) If the interface of R3 to network 198.1.34.0/24 fails, what is the resulting IP routing table of R4?

OSPF (Open Shortest Path First) Protocol

- OSPF is a *link-state* routing protocol
  - Each router learns and maintains a database with the complete network topology view, known as the **Link-State Database (LSDB)**.
  - Each router sends its part of the network topology to all routers in the form of **Link-State Advertisements (LSAs)**:
    - LSAs are sent by their responsible routers through flooding to all other routers.
    - LSAs are sent on bootstrap (i.e., when they start) or when the topology changes.
- Each router determines the entries of its IP routing table for each remote IP network based on the LSDB:
  - The router uses its LSDB to calculate the minimum cost routing paths from it towards each remote IP network (using Dijkstra's algorithm).
  - The router adds to its IP routing table an entry towards each next-hop (i.e., neighbour) router providing the minimum cost.

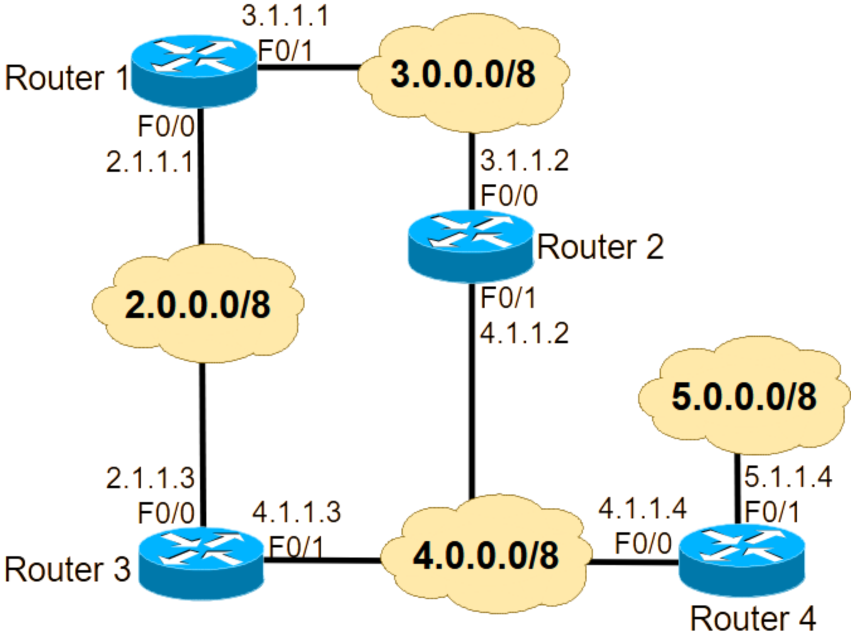
OSPF Path Costs

- Each router interface has an associated OSPF cost.
  - Cost values are administratively configured by the manager
  - Cost values are coded in 2 bytes (can be between 1 and 65535)
- The cost of a routing path from a router to a remote IP is the sum of the cost values of the output interfaces of each router along the path.



- Administrators can use cost values to control the routing paths (illustration in the next slide).

Routing Tables with OSPF



OSPF costs = 10 (in all interfaces of all routers)

```
Router 1
C   2.0.0.0/8 is directly connected, F0/0
C   3.0.0.0/8 is directly connected, F0/1
O   4.0.0.0/8 [110/20] via 3.1.1.2, 00:01:13, F0/1
              [110/20] via 2.1.1.3, 00:01:31, F0/0
O   5.0.0.0/8 [110/30] via 3.1.1.2, 00:01:10, F0/1
              [110/30] via 2.1.1.3, 00:01:10, F0/0
```

After a failure of interface F0/1 in Router 2:

```
Router 1
C   2.0.0.0/8 is directly connected, F0/0
C   3.0.0.0/8 is directly connected, F0/1
O   4.0.0.0/8 [110/20] via 2.1.1.3, 00:01:18, F0/0
O   5.0.0.0/8 [110/30] via 2.1.1.3, 00:01:00, F0/0

Router 2
O   2.0.0.0/8 [110/20] via 3.1.1.1, 00:01:13, F0/0
C   3.0.0.0/8 is directly connected, F0/0
O   4.0.0.0/8 [110/30] via 3.1.1.1, 00:01:13, F0/0
O   5.0.0.0/8 [110/40] via 3.1.1.1, 00:01:10, F0/0
```

After configuration of interface F0/1 in Router 2 with a OSPF cost = 5:

```
Router 1
C   2.0.0.0/8 is directly connected, F0/0
C   3.0.0.0/8 is directly connected, F0/1
O   4.0.0.0/8 [110/15] via 3.1.1.2, 00:01:13, F0/1
O   5.0.0.0/8 [110/25] via 3.1.1.2, 00:01:10, F0/1
```

OSPF Protocol Principles

- OSPF must guarantee that the LSDB is synchronized in all routers
  - Otherwise, the routing tables on the different OSPF routers can become inconsistent between them
- For that:
  1. Each router must have a unique Router ID (RID)
  2. Each network (stub or transit) must have a unique Network ID (NID)
  3. Each OSPF router must reliably know what are its OSPF neighbour routers on each of its interfaces (OSPF adjacencies)
  4. The protocol must guarantee that when new LSAs are sent by a OSPF router through an interface, they are effectively received by all OSPF neighbour routers on that interface.
- Router IDs and Network IDs are in the form of IP addresses
- OSPF runs directly over IP
- OSPF uses different message types in tasks 3. and 4. (seen later)

OSPF Link-State Database (LSDB)

- The OSPF LSDB contains at least 2 types of entries:
  - Router Link States
    - There is one Router Link State entry per existing OSPF router
    - The Router Link State entry is identified by the RID (Router ID) of the router.
    - The Router Link State of a given router identifies the networks it is directly connected to and the associated OSPF costs
  - Net Link States – Transit networks related information table
    - There is one Net Link State entry per existing IP transit network
    - The Network Link State entry is identified by the NID (Network ID) of the network.
    - The Net Link State of a given transit network identifies the routers it is directly connected to.
- Each entry (either Router or Net Link State) has an associated advertiser router, i.e., the router responsible to send the entry (through flooding) in the form of a LSA to all other routers

OSPF Router ID (RID)

- The Router ID (RID) is:
  - (by default) the highest IPv4 address among all router interfaces at the instant of the OSPF activation, or
  - an IP address configured by the administrator (the administrator must guarantee that the IDs of all routers are unique).
- When the IP address of a router interface is used as its RID (the default method):
  - if the physical interface fails and the router (or OSPF process) restarts, the RID changes;
  - this RID change requires the flood of new LSAs and makes it more difficult for network administrators to troubleshoot and manage OSPF;
  - administratively configured RIDs guarantee that they remain the same, regardless of the state of the physical interfaces.
- <u>Each router is the advertising router of its Router Link State.</u>

OSPF Adjacencies

- Each router first establishes neighbour adjacencies on each LAN through the Hello protocol.
- Each router sends/receives Hello messages to/from its neighbouring routers on each interface running OSPF.
  - The destination address is the all-OSPF-routers multicast address 224.0.0.5.
- The routers exchange Hello messages subject to protocol-specific parameters, such as checking whether the neighbour is in the same area, using the same hello interval, and so on.
  - Routers declare that the neighbour is up when the exchange is complete.
- On each LAN, routers use Hello messages to elect one router as the Designated Router (DR) and another as the Backup Designated Router (BDR) – next 2 slides.

DR (Designated Router) and BDR (Backup Designated Router) Election

- On each LAN:
  - The first router to activate the Hello protocol becomes the DR.
  - The second router to activate the Hello protocol becomes the BDR.
- If multiple routers activate the Hello protocol simultaneously:
  - The DR and the BDR are the routers with the highest and second highest priorities, respectively.
  - By default, all routers have the same priority, but different priority values can be administratively configured.
  - In case of tie, the DR and the BDR are the routers with the highest and second highest RIDs (Router IDs), respectively.

DR and BDR Election (continuation)

- When the DR fails, the BDR assumes the role of DR.
  - The BDR does not perform any DR functions when the DR is active.
  - The choice of the new BDR is again the router with the highest priority or the highest RID in case of equal priorities.
- After the election, the DR and BDR maintain that role, independently of which routers join the OSPF process afterwards.
- The advertising router of the Net Link State associated with the IP transit network is its DR.
- The NID (Network ID) of the Net Link State is the IP address of the DR interface connected to the network.

OSPF Link-State Database (LSDB)

Visualization of a LSDB in a router:

```
           OSPF Router with ID (20.20.20.1) (Process ID 1)

                     Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
20.20.20.1      20.20.20.1      40          0x8000000A 0x00E7FB 2
30.30.30.2      30.30.30.2      69          0x80000006 0x002906 2
30.30.30.3      30.30.30.3      41          0x80000007 0x00283D 2

                     Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
10.10.10.3      30.30.30.3      41          0x80000001 0x00051C
20.20.20.2      30.30.30.2      70          0x80000001 0x00A164
30.30.30.3      30.30.30.3      154         0x80000001 0x00A91C
```

- The visualization is being done in router with ID 20.20.20.1
- There are 3 Router Link State entries:
  - The Link State ID is always the ID of the advertising router
- There are 3 Net Link State entries:
  - The advertising router is always one of the existing routers
  - The Link State ID is the IP address of the DR

OSPF LSDB: Router Link State entry

- The Router Link State of a router contains the information about the networks directly connected to the router.
  - In Stub networks, the Link ID is the IP network address, and the Link Data is the network mask

```
LS age: 321
Options: (No TOS-capability, DC)
LS Type: Router Links
Link State ID: 20.20.20.1
Advertising Router: 20.20.20.1
LS Seq Number: 8000000A
Checksum: 0xE7FB
Length: 48
Number of Links: 2

  Link connected to: a Transit Network
   (Link ID) Designated Router address: 20.20.20.2
   (Link Data) Router Interface address: 20.20.20.1
    Number of TOS metrics: 0
     TOS 0 Metrics: 1

  Link connected to: a Transit Network
   (Link ID) Designated Router address: 10.10.10.3
   (Link Data) Router Interface address: 10.10.10.1
    Number of TOS metrics: 0
     TOS 0 Metrics: 1
```

RID (Router ID)

Number of connected IP networks

Network Type (Transit of Stub)

OSPF interface cost

NID (Network ID)

IP address of the interface

OSPF LSDB: Net Link State entry

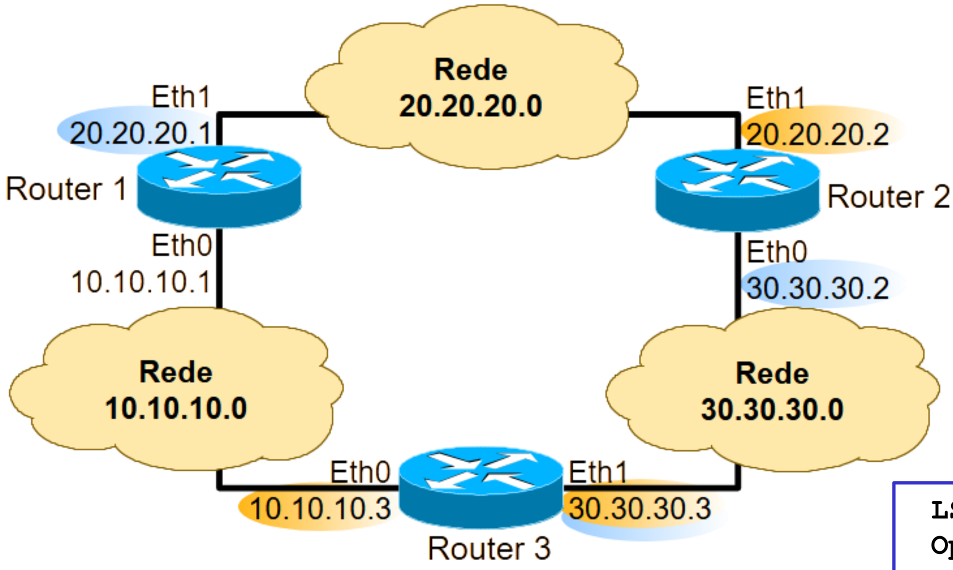
- The Net Link State of a transit network identifies the routers directly connected to the network.
  - Stub networks do not have Net Link State entries.

```
LS age: 483
Options: (No TOS-capability, DC)
LS Type: Network Links
Link State ID: 10.10.10.3 (address of Designated Router)
Advertising Router: 30.30.30.3
LS Seq Number: 80000001
Checksum: 0x51C
Length: 32
Network Mask: /24
          Attached Router: 30.30.30.3
          Attached Router: 20.20.20.1
```

NID (Network ID)

List of RIDs of all routers connected to the IP network

OSPF LSDB Example



Router Link State of router 20.20.20.1:

```
LS age: 321
Options: (No TOS-capability, DC)
LS Type: Router Links
Link State ID: 20.20.20.1
Advertising Router: 20.20.20.1
LS Seq Number: 8000000A
Checksum: 0xE7FB
Length: 48
Number of Links: 2

  Link connected to: a Transit Network
   (Link ID) Designated Router address: 20.20.20.2
   (Link Data) Router Interface address: 20.20.20.1
    Number of TOS metrics: 0
     TOS 0 Metrics: 1

  Link connected to: a Transit Network
   (Link ID) Designated Router address: 10.10.10.3
   (Link Data) Router Interface address: 10.10.10.1
    Number of TOS metrics: 0
     TOS 0 Metrics: 1
```

Net Link State of network 10.10.10.3:

```
LS age: 483
Options: (No TOS-capability, DC)
LS Type: Network Links
Link State ID: 10.10.10.3 (address of …
Advertising Router: 30.30.30.3
LS Seq Number: 80000001
Checksum: 0x51C
Length: 32
Network Mask: /24
          Attached Router: 30.30.30.3
          Attached Router: 20.20.20.1
```

OSPF Messages

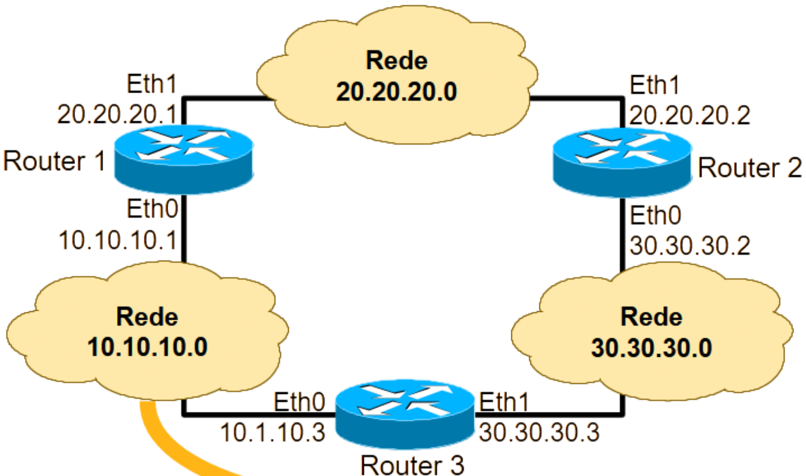
- Building adjacencies with neighbour routers:
  - Hello protocol
  - For reliability, the Hello messages sent by each router include the Router ID of the neighbour routers
    - In this way, each router knowns it is known by all neighbour routers
- Learning and maintaining the same LSDB in all routers:
  - Database Description (DBD) - Send a summary of the current LSDB.
  - Link-State Request (LSR) - Request specific LSAs from another router.
  - Link-State Update (LSU) - Send specifically LSAs.
  - Link-State Acknowledgement (LSAck) - Acknowledge the reception of LSUs.
  - LSAck messages provide reliable LSA transfer
    - If the sender of a LSU does not receive a LSAck from the receiver, it repeats sending the LSU until the LSAck is received

OSPF Example

OSPF activated on Router 1

OSPF activated on Router 3

OSPF activated on Router 2



Time	Source	Destination	Protocol	Info
0.000000	10.10.10.1	224.0.0.5	OSPF	Hello Packet
10.002318	10.10.10.1	224.0.0.5	OSPF	Hello Packet
20.003116	10.10.10.1	224.0.0.5	OSPF	Hello Packet

80.000000	10.10.10.3	224.0.0.5	OSPF	Hello Packet
83.683033	10.10.10.3	224.0.0.5	OSPF	LS Update
83.715683	10.10.10.3	224.0.0.5	OSPF	Hello Packet
83.717864	10.10.10.1	10.10.10.3	OSPF	Hello Packet
83.726166	10.10.10.3	10.10.10.1	OSPF	DB Descr.
83.726258	10.10.10.3	10.10.10.1	OSPF	Hello Packet
83.728433	10.10.10.1	10.10.10.3	OSPF	DB Descr.
83.732590	10.10.10.3	10.10.10.1	OSPF	DB Descr.
83.734733	10.10.10.1	10.10.10.3	OSPF	DB Descr.
83.738942	10.10.10.3	10.10.10.1	OSPF	LS Request
83.741083	10.10.10.1	10.10.10.3	OSPF	LS Update
84.240362	10.10.10.3	224.0.0.5	OSPF	LS Update
86.245792	10.10.10.3	224.0.0.5	OSPF	LS Acknowledge
86.380876	10.10.10.1	224.0.0.5	OSPF	Hello Packet
86.741036	10.10.10.1	224.0.0.5	OSPF	LS Acknowledge
93.721376	10.10.10.3	224.0.0.5	OSPF	Hello Packet
96.380005	10.10.10.1	224.0.0.5	OSPF	Hello Packet

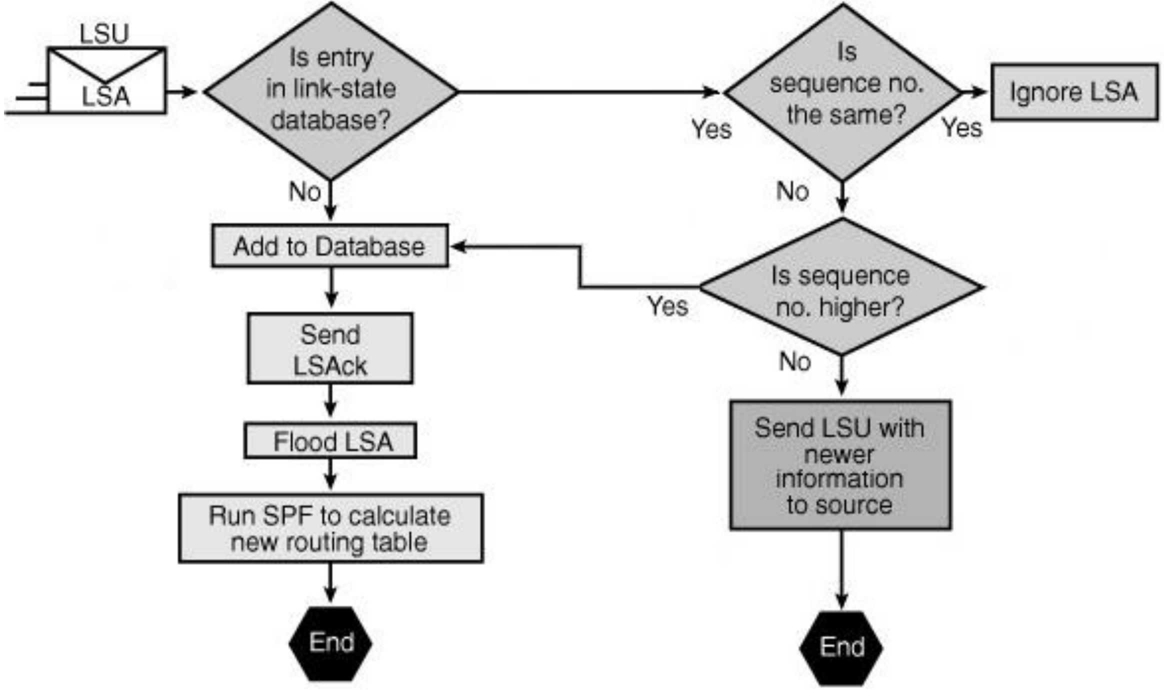
213.780338	10.10.10.3	224.0.0.5	OSPF	Hello Packet
216.542473	10.10.10.1	224.0.0.5	OSPF	Hello Packet
216.568852	10.10.10.1	224.0.0.5	OSPF	LS Update
217.048427	10.10.10.1	224.0.0.5	OSPF	LS Update
217.084909	10.10.10.1	224.0.0.5	OSPF	LS Update
219.067748	10.10.10.3	224.0.0.5	OSPF	LS Acknowledge
219.650308	10.10.10.1	224.0.0.5	OSPF	LS Update
222.150349	10.10.10.3	224.0.0.5	OSPF	LS Acknowledge
223.779492	10.10.10.3	224.0.0.5	OSPF	Hello Packet
224.284149	10.10.10.3	224.0.0.5	OSPF	LS Update
224.789598	10.10.10.1	224.0.0.5	OSPF	LS Update
224.789775	10.10.10.3	224.0.0.5	OSPF	LS Update
226.545718	10.10.10.1	224.0.0.5	OSPF	Hello Packet
226.785254	10.10.10.1	224.0.0.5	OSPF	LS Acknowledge
227.294756	10.10.10.3	224.0.0.5	OSPF	LS Acknowledge
233.779863	10.10.10.3	224.0.0.5	OSPF	Hello Packet
236.544658	10.10.10.1	224.0.0.5	OSPF	Hello Packet

OSPF: flooding process on each LAN

1. When a router notices a change in a link state of its responsibility (either its Router Link State or the Net Link State of the networks that it is the Designated Router) it sends an LSU message, which includes the updated LSA entry with the sequence number incremented, to 224.0.0.6 (the DR multicast address).
   - This message is received by the DR and the BDR.
   - An LSU message might contain several distinct LSAs.
2. The DR receives the LSU, processes it, acknowledges the receipt of the change and floods the LSU to all routers on the LAN using the OSPF multicast address 224.0.0.5.
3. After receiving the LSU, each router responds to the DR with an LSAck.
   - For reliability, each LSA is acknowledged separately.
4. Each router updates its LSDB using the LSU that includes the changed LSA and updates the IP routing table if necessary.

5. If it is the DR flooding the LSU to the LAN, the process starts on step 2.

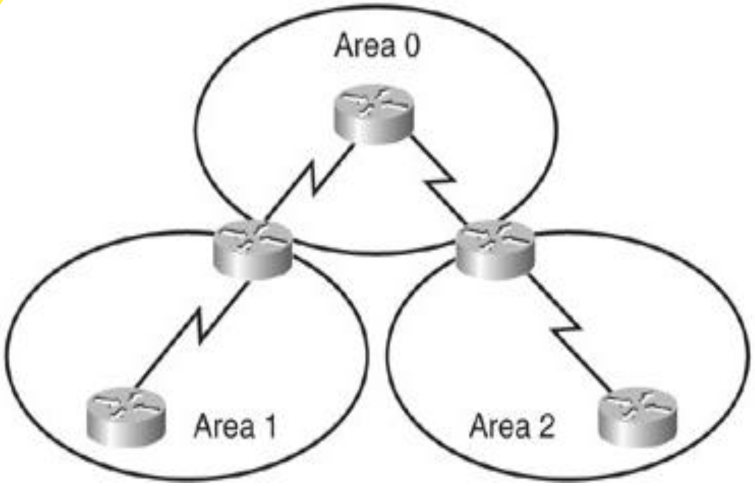
OSPF: processing incoming LSUs



- When a router receives an LSU from one incoming interface:
  - If the LSA entry does not exist, the router adds the entry to its LSDB, sends back a LSAck and floods the information to the other interfaces.
  - If the entry exists and the received LSA has the same sequence number, the router ignores the LSA entry.
  - If the entry exists and the LSA contains newer information (i.e., it has a higher sequence number), the router adds the entry to its LSDB, sends back an LSAck and floods the information to the other interfaces.
  - If the entry already exists and the LSA contains older information (i.e., it has a lower sequence number), it sends an LSU to the sender with its newer information.

OSPF Hierarchical Routing (I)

- In large networks, the resulting topology is highly complex, and the number of potential paths to each destination is large.
- Shortest path calculations can be very complex and can take significant time due to:
  - **Large LSDB** - Because the LSDB covers the topology of the entire network, each router must maintain an entry for every router and every transit network.
  - **Frequent shortest path calculations** - In a large network, changes are inevitable; so, the routers spend many CPU cycles recalculating the shortest paths and updating the IP routing table when a change occurs.
  - **Large IP routing tables** - OSPF does not perform route summarization by default. If the routes are not summarized, the routing tables include one entry per existing IP network and, therefore, can become very large.
- Link-state routing protocols usually reduce the complexity of the shortest path calculations by partitioning the network into areas.

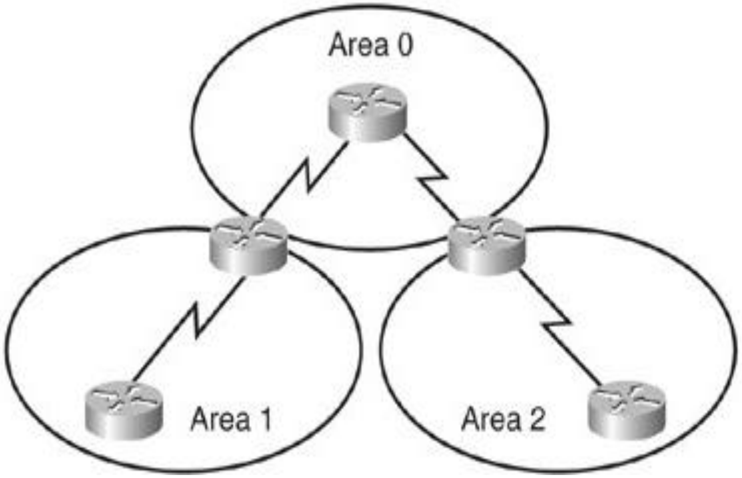


OSPF Hierarchical Routing (II)

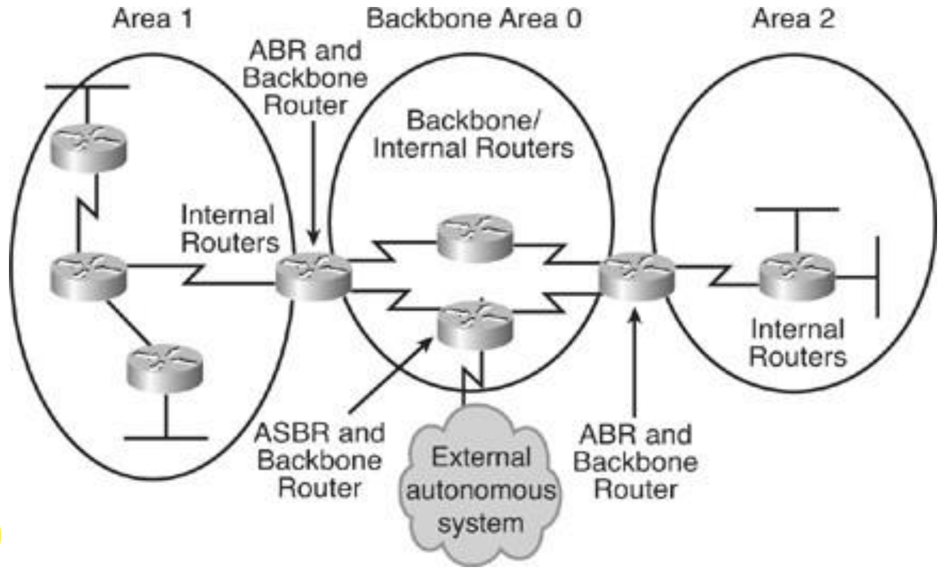
- Using multiple OSPF areas has several important advantages:
  - Reduced frequency of shortest path calculations.
    - Detailed route information only exists within each area
    - It is not necessary to flood all link-state changes to all other areas.
    - Only routers that are affected by the change need to recalculate the shortest paths and the impact of the change is localized within the area.
  - Reduced updates overhead.
    - Rather than send an update about each router and network within an area, a router can advertise a single summarized route or a small number of routes between areas, thereby reducing the overhead associated with updates when they cross areas.
  - Smaller routing tables.
    - Routers can be configured to summarize the routes into one or more summary network addresses.
    - Advertising these summaries reduces the number of messages propagated between areas but keeps all networks reachable.

OSPF Two-Layer Area Hierarchy

- Backbone area (Area 0)
  - The OSPF area whose primary function is the fast and efficient forwarding of IP packets between the other areas (or from/to other networks outside the Autonomous System).
  - Generally, end users are not attached to the networks of the backbone area.
  - Hierarchical networking defines Area 0 as the core to which all other areas connect.
- Regular (non backbone) area
  - An OSPF area whose primary function is to connect client users and resources to the Autonomous System.
  - Regular areas are usually set up along functional or geographic groupings.
  - By default, a regular area cannot be used as transit between areas (IP packet between different regular areas are routed through the backbone area.

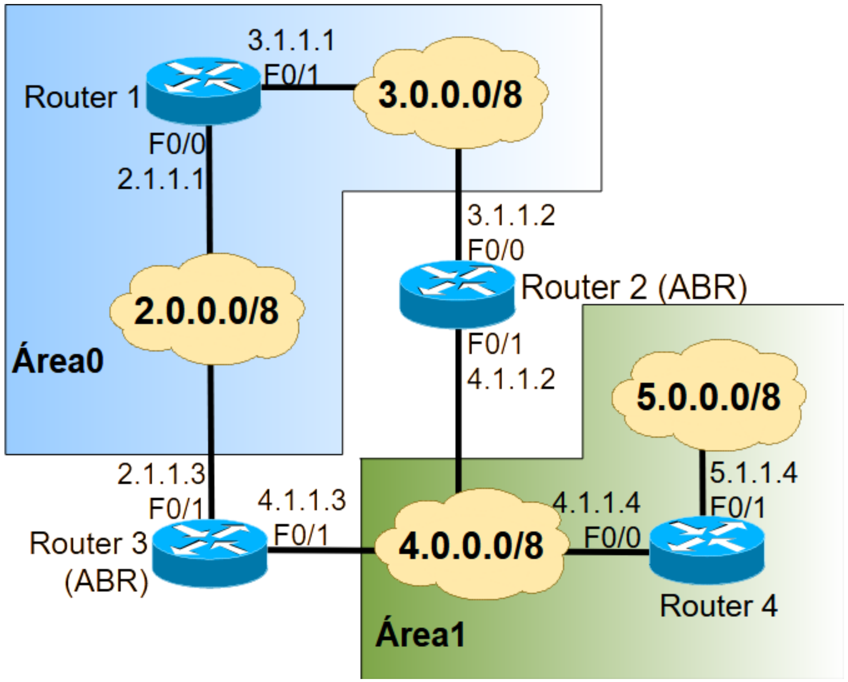


OSPF Router Types



- Internal router
  - Router with all interfaces connected to the same area
- Area Border Router (ABR)
  - Router with interfaces connected to multiple areas
  - ABRs maintain a separate LSDB for each connected area.
  - ABRs connect Area 0 to a non backbone area and are exit points for the area
- Autonomous System Boundary Router (ASBR)
  - Router with at least one interface connected to a different routing domain (another Autonomous System or a domain using another routing protocol).
  - ASBRs can redistribute external routes into the OSPF domain and vice versa.
- A router can be simultaneously an ABR and an ASBR
  - For example, if a router is connected to Area 0 and Area 1, and connected to a different routing domain.

OSPF Hierarchical Routing Example



OSPF costs = 10
(in all interfaces of all routers)

Advertising Router: 3.1.1.1	Advertising Router: 4.1.1.2
Number of Links: 2	Number of Links: 1
Router Interface address: 3.1.1.1	Router Interface address: 3.1.1.2
TOS 0 Metrics: 10	TOS 0 Metrics: 10
Router Interface address: 2.1.1.1	Advertising Router: 4.1.1.3
TOS 0 Metrics: 10	Number of Links: 1
	Router Interface address: 2.1.1.3
	TOS 0 Metrics: 10

Router Link States from Router 1

Link State ID: 4.0.0.0	Link State ID: 5.0.0.0
Advertising Router: 4.1.1.2	Advertising Router: 4.1.1.2
Network Mask: /8	Network Mask: /8
TOS: 0 Metric: 10	TOS: 0 Metric: 20
Link State ID: 4.0.0.0	Link State ID: 5.0.0.0
Advertising Router: 4.1.1.3	Advertising Router: 4.1.1.3
Network Mask: /8	Network Mask: /8
TOS: 0 Metric: 10	TOS: 0 Metric: 20

Summary Net Link States from Router 1

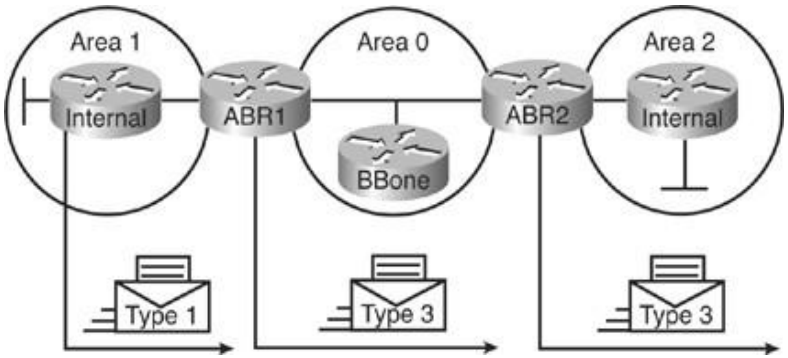
Link State ID: 2.1.1.3	Link State ID: 3.1.1.2
Network Mask: /8	Network Mask: /8
Attached Router: 3.1.1.1	Attached Router: 3.1.1.1
Attached Router: 4.1.1.3	Attached Router: 4.1.1.2

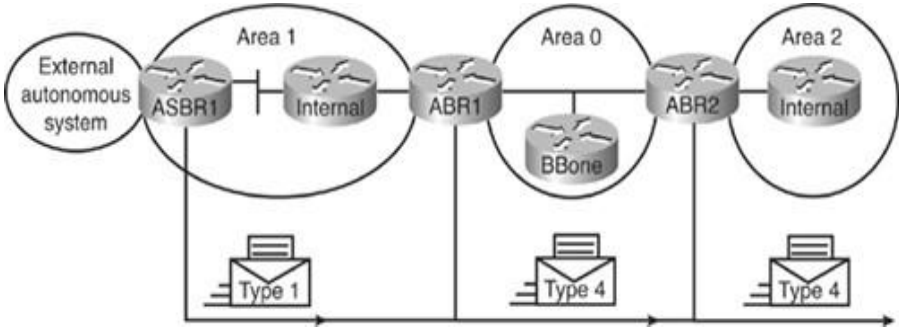
Net Link States from Router 1

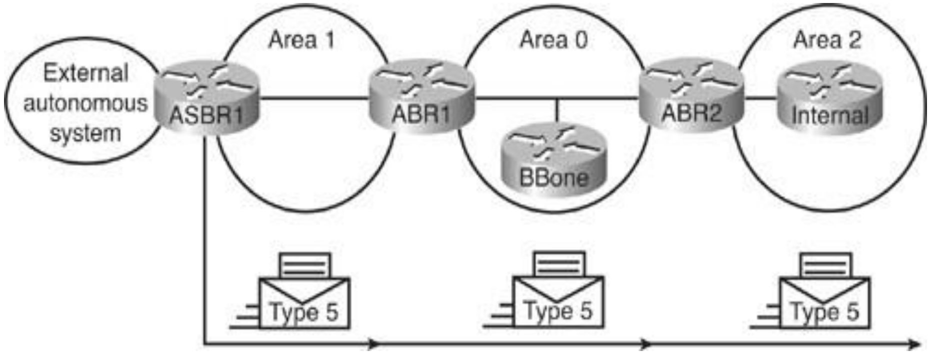
Another LSA type: Summary Net Link State

OSPF relevant LSA types

- Type 1 (Router LSA) - Routers generate their LSAs for each area to which they belong.
- Type 2 (Network LSA) - DRs generate network LSAs for their transit networks.
- Types 3 and 4 (Summary LSA) - ABRs generate summary net link states:
  - Type 3 describes routes to the networks of the other areas.
  - Type 4 describes routes to existing ASBRs.
- Type 5 (AS External LSA) - ASBRs generate Autonomous System external link advertisements. External link advertisements describe routes to destinations external to the autonomous system.



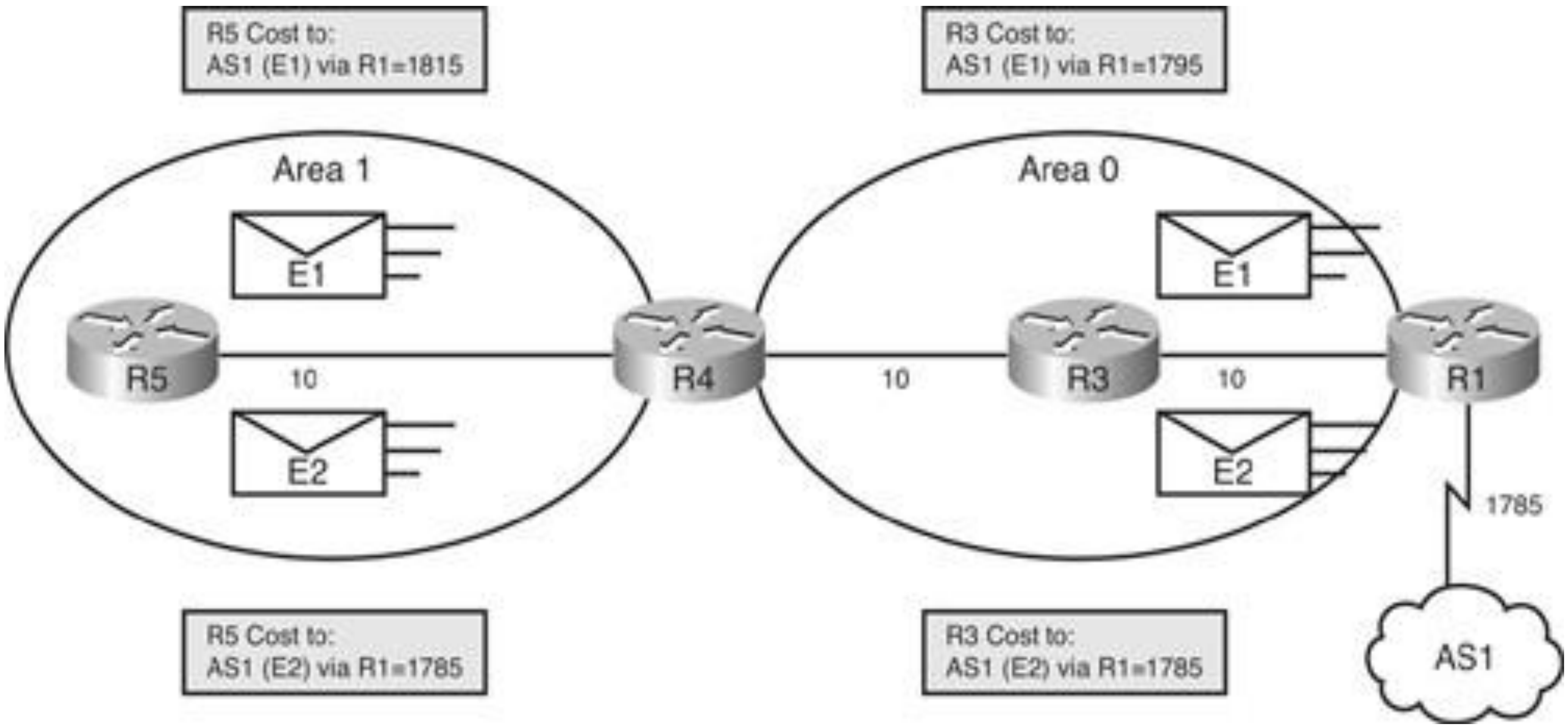




Types of OSPF Routes

- OSPF intra-area (router LSAs and net LSAs)
  - Networks in one of the areas of the router (advertised by Router and Net LSAs).
- Inter-area (summary LSAs)
  - Networks outside the router areas of the router but within the OSPF Autonomous System (advertised by Summary LSAs).
- Type 2 external routes (E2)
  - Networks outside the OSPF domain (advertised by External LSAs).
  - The cost of OSPF E2 routes is always the announced external cost.
  - Use this type if only one ASBR is advertising an external route.
- Type 1 external routes (E1)
  - Networks outside the OSPF domain (advertised by External LSAs).
  - Routers calculate the path cost by adding the announced external cost to the internal minimum path cost.
  - Use this type when multiple ASBRs are advertising the same outside networks to obtain optimal routing).
  - Always preferred over Type 2 external routes (E2).

OSPF route path cost illustration with external routes of Type E1 and E2



OSPF version 3 (OSPFv3) Protocol

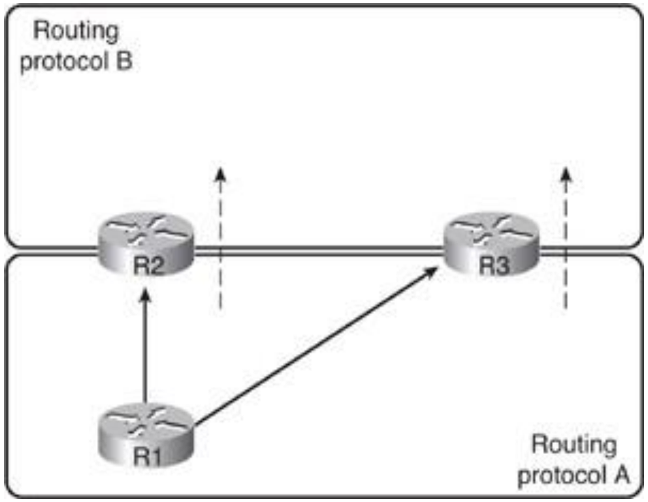
- OSPFv3 is a routing protocol for IPv6 networks
- It is based on OSPFv2, with modifications:
  - OSPFv3 runs over IPv6
  - It runs over links (through link-local IPv6 network addresses) instead of IPv6 subnet
    - It might have multiple IPv6 instances per link whose prefixes are distributed
    - Uses multicast group addresses FF02::5 (OSPF IGP) and FF02::6 (OSPF IGP Designated Routers)
  - Topology representation is not IPv6-specific
    - Router ID, Area ID, Link ID remain a 4 bytes IPv4 format
    - Neighbour routers are always identified by Router ID (4 bytes)
    - With an additional table with mapping between IPv6 prefixes and Link IDs

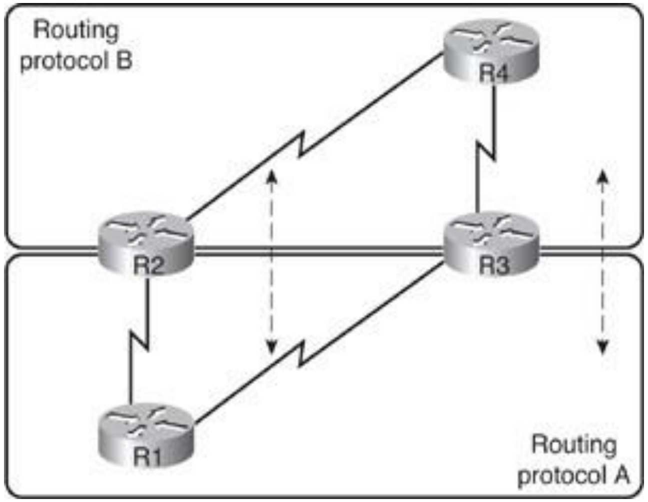
OSPFv3 – New LSA Types and Flooding Scopes

- New LSA types:
  - Link LSA (Type 8)
    - Informs neighbour routers of link local address
    - Informs neighbour routers of IPv6 prefixes on link
  - Intra-Area Prefix LSA (Type 9)
    - Associates IPv6 prefixes to links (announced by the Designated Router of each link)
- Flooding scope for LSAs has been generalized:
  - Three flooding scopes for LSAs
    - Link-local
    - Area
    - AS
  - LSA Type encoding expanded to 16 bits
    - Includes flooding scope

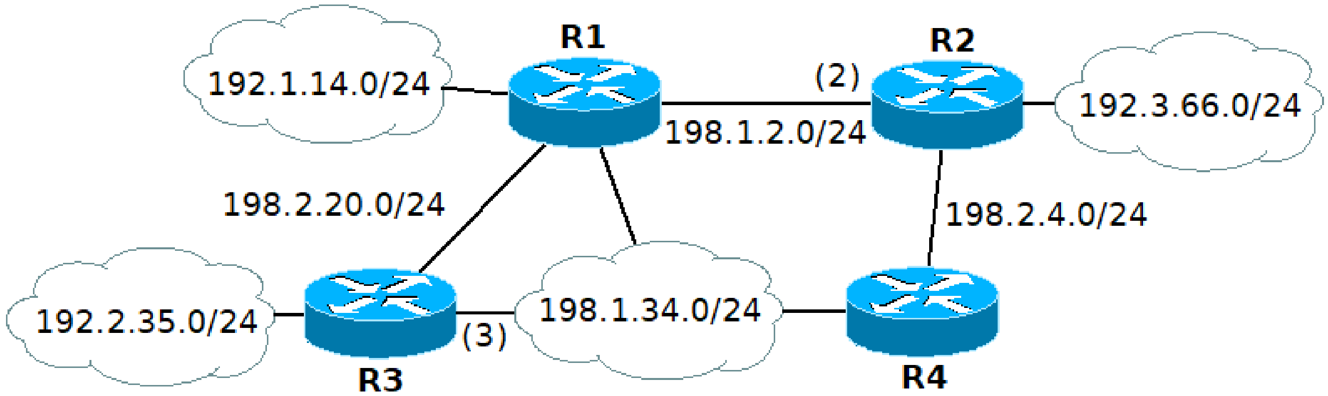
Route Redistribution

- Domains with difference routing protocols can exchange routes.
- This is called route redistribution.
  - One-way redistribution → Redistributes only the networks learned from one routing protocol into the other routing protocol.
    - Uses a default or static route so that devices in that other part of the network can reach the first part of the network
  - Two-way redistribution → Redistributes routes between the two routing processes in both directions
- Prevent routing loops when there are multiple boundary routers.
- Static routes can also be redistributed.





Example 2 (OSPF)



Consider the IP network of the figure (the host part of each IP address is the number of the router's name). All routers except R3 are configured with OSPF in all IP networks. R3 is configured with OSPF in networks 192.2.20.0/24 and 192.1.34.0/24. R2 is announcing 192.2.35.0/24 with an external route of type E1 and external cost 5. The OSPF costs that are not equal to 1 are defined in the figure between brackets.

a) What are the OSPF entries of the IP routing table of R3?

b) What are the OSPF entries of the IP routing table of R2?

c) What is the sequence of routers crossed by the ICMP Echo Requests and ICMP Echo Replies in a ping from R4 to a host in 192.2.35.0/24.

d) If the link with network address 198.2.20.0/24 fails, what is the resulting IP routing table of R2?